

2nd Session: NLP meets GenAI :LLMs & RAG

27 / 12 / 2024



Hello everynryan



MOHAMED ARBI NSIBI

- Final year ICT engineering student@ SUP'COM
- GDG Carthage member
- Mentor of GDGoC SUP'COM & ISAMM
- Former GDSC Lead 23/24



<https://huggingface.co/Goodnight7>



<https://www.linkedin.com/in/mohammed-arbi-nsibi-584a43241/>



mohammedarbinsibi@gmail.com

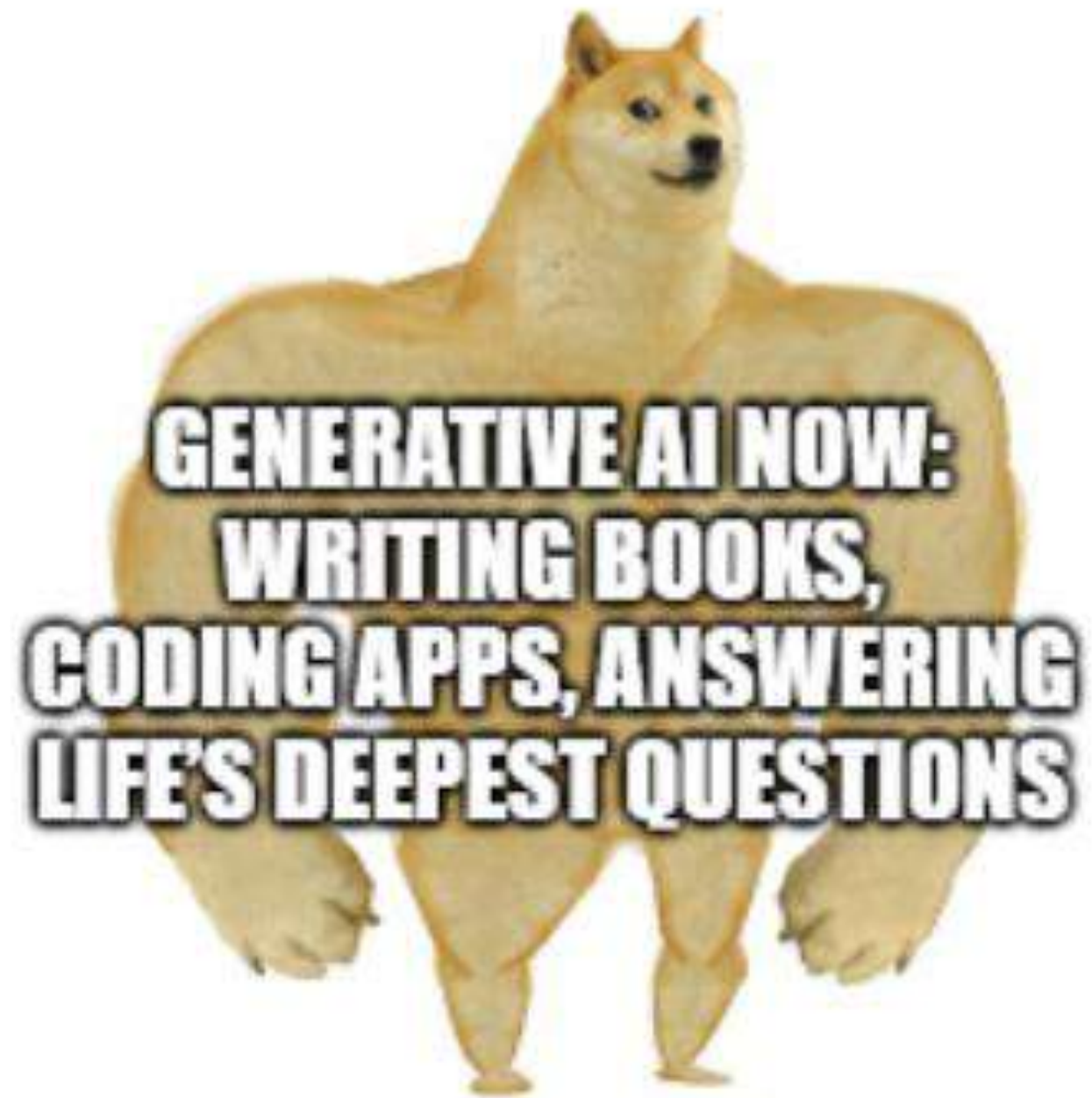
Content

- NLP
 - Pipeline
- LLMs
 - Transformers
 - Let's code
- Why we need RAG
- Speaking on Your Behalf : Building a ChatBot
- QUIZ



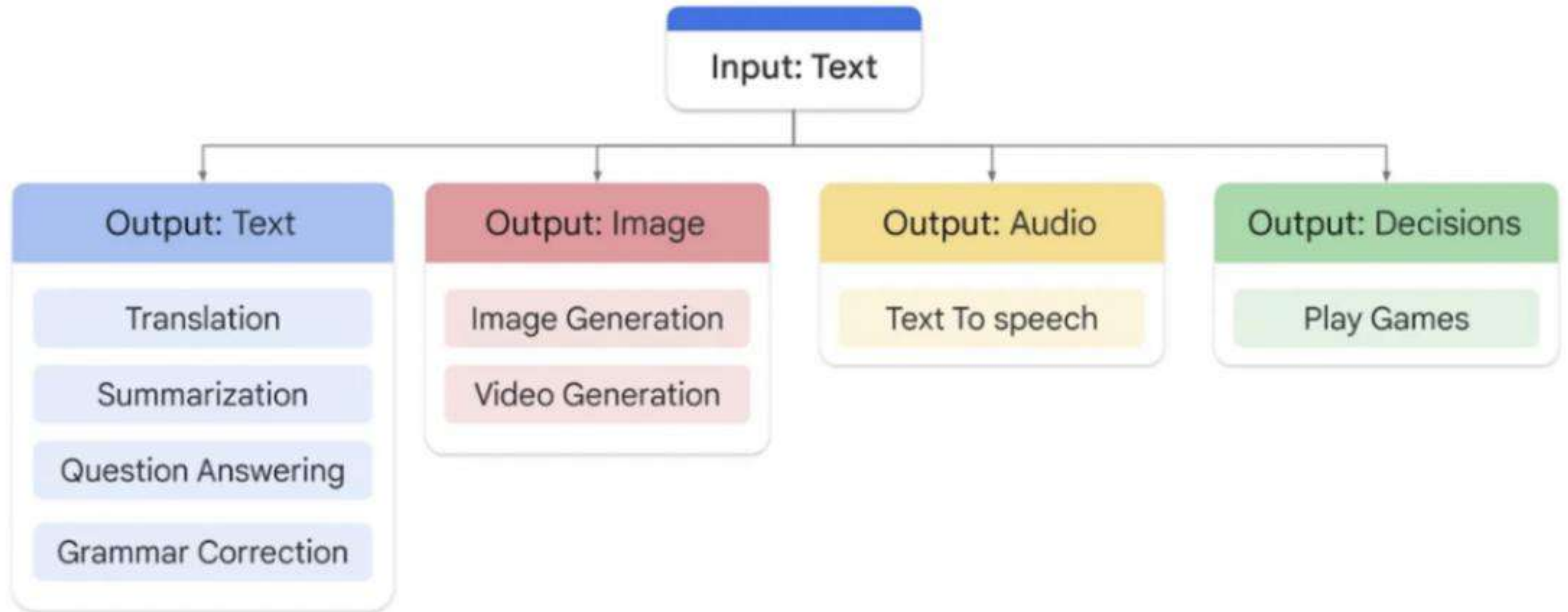
Google Developer Groups
On Campus • SUP'COM

By Mohammed Arbi Nsibi

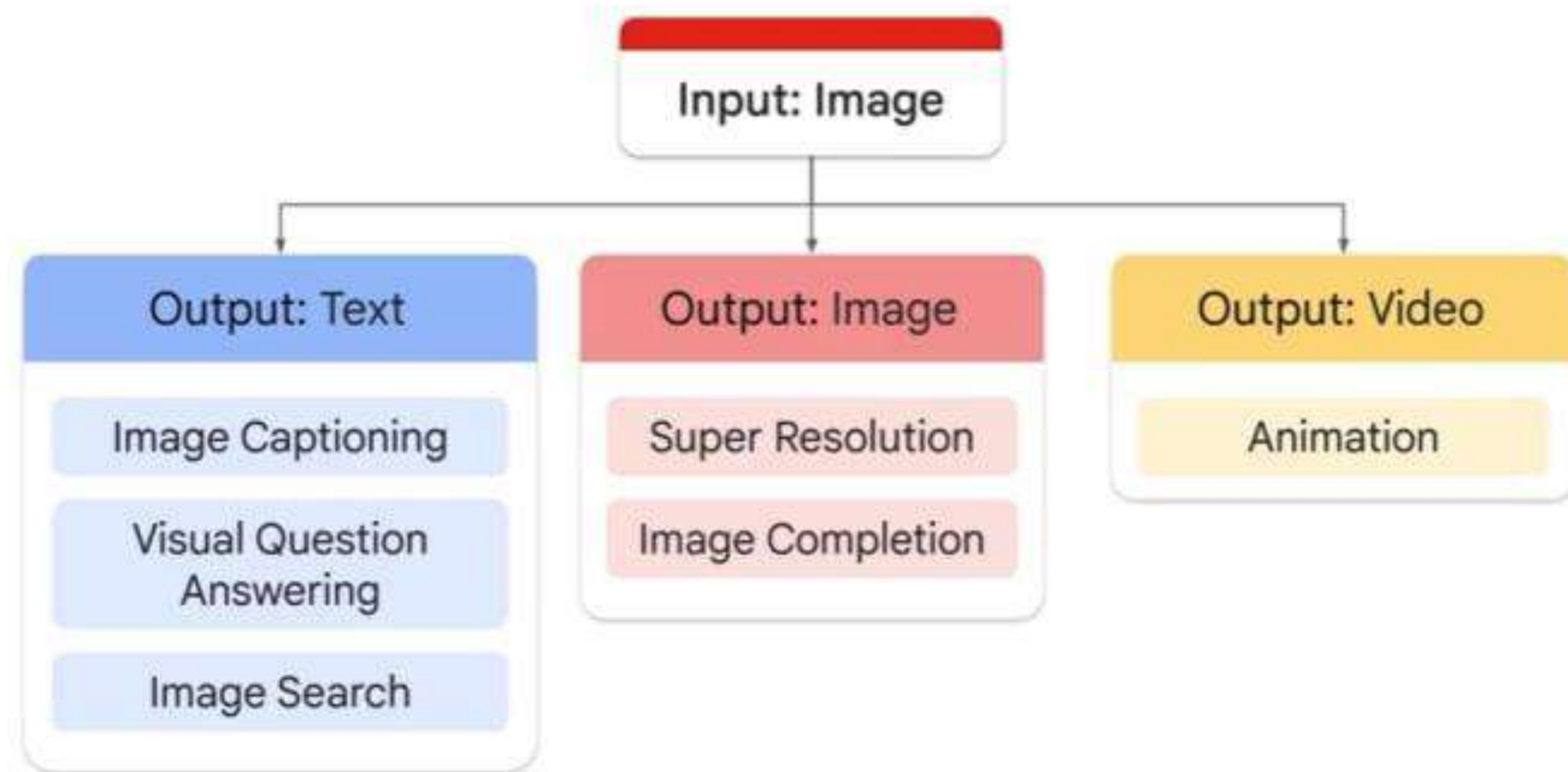


By Mohammed Arbi Nsibi

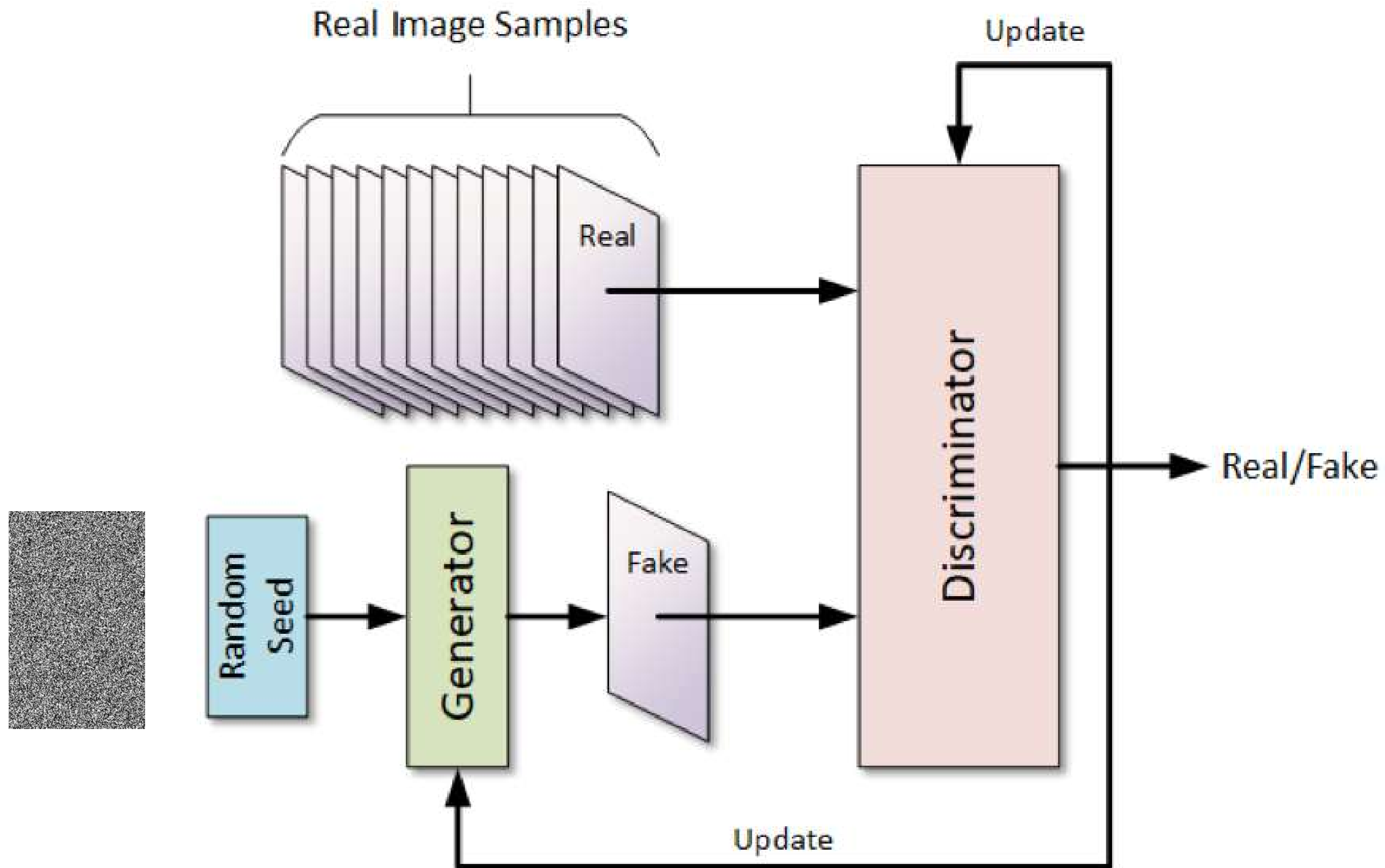
Types of Generative AI Based on Data



Types of Generative AI Based on Data



GANs: Generative adversarial network



Diffusion models :

DALL-E

DALL-E ▾



A wolf running through the woods



Here are two depictions of the wolf running through the woods. Let me know if you'd like any adjustments!

Can you make it more wintry?



Here are the updated images with a more wintry atmosphere, featuring the wolf running through a snowy forest. Let me know what you think or if you'd like any further adjustments!

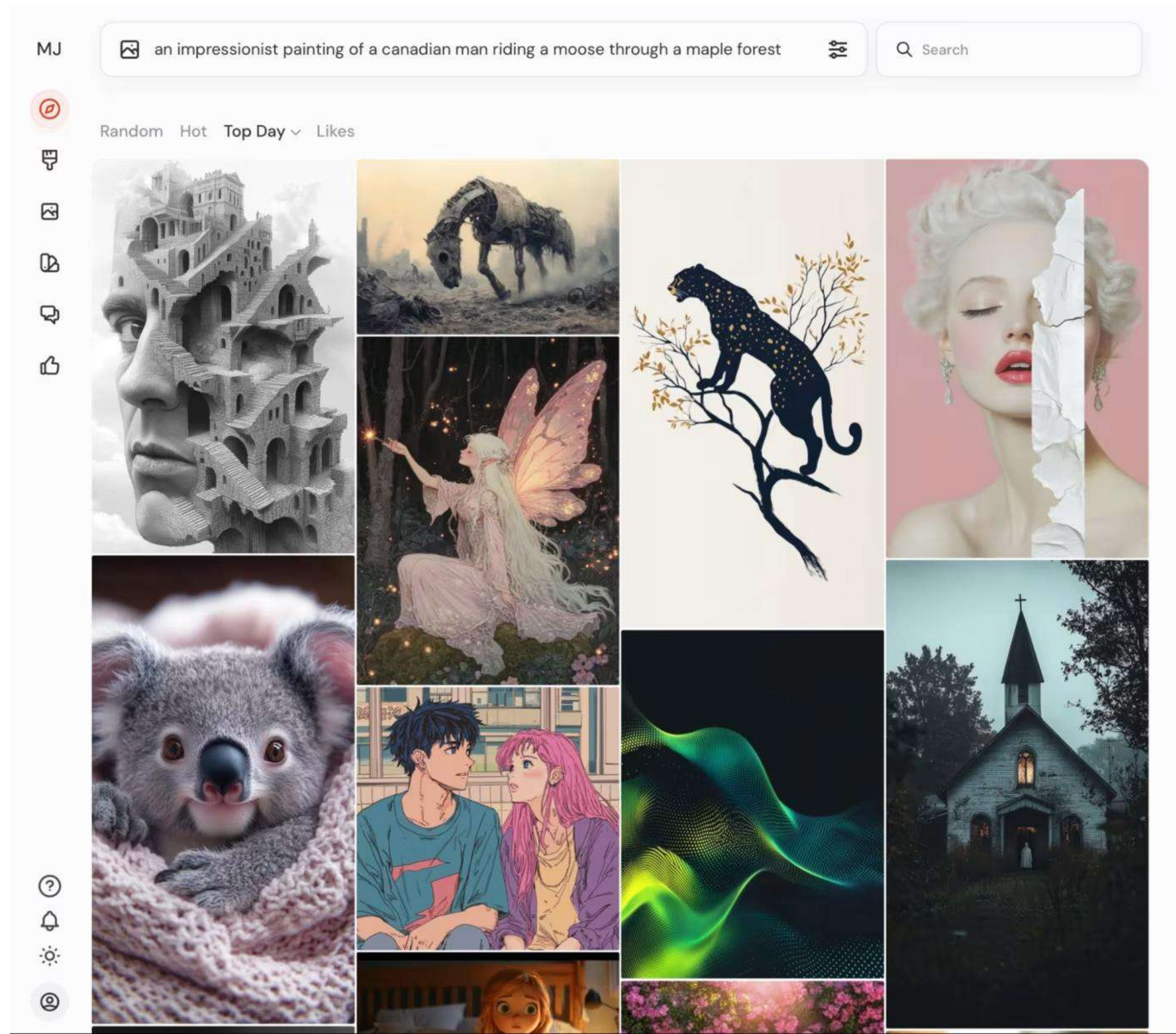


<https://openai.com/index/dall-e-2/>

Diffusion models :

Midjourney

<https://www.midjourney.com/home>



ChatGPT



Explain UX design like you would to a 2-year-old



UX design is like making a toy for a big kid. Just like how a toy needs to be fun and easy for a little kid to play with, a website or app needs to be easy for people to use. We make sure that things are in the right place and that the colors look pretty. We also make sure

ChatGPT (Dec 23 Version) - Free Research Preview. Our goal is to make AI systems more natural and safe to interact with. Your feedback will help us improve.

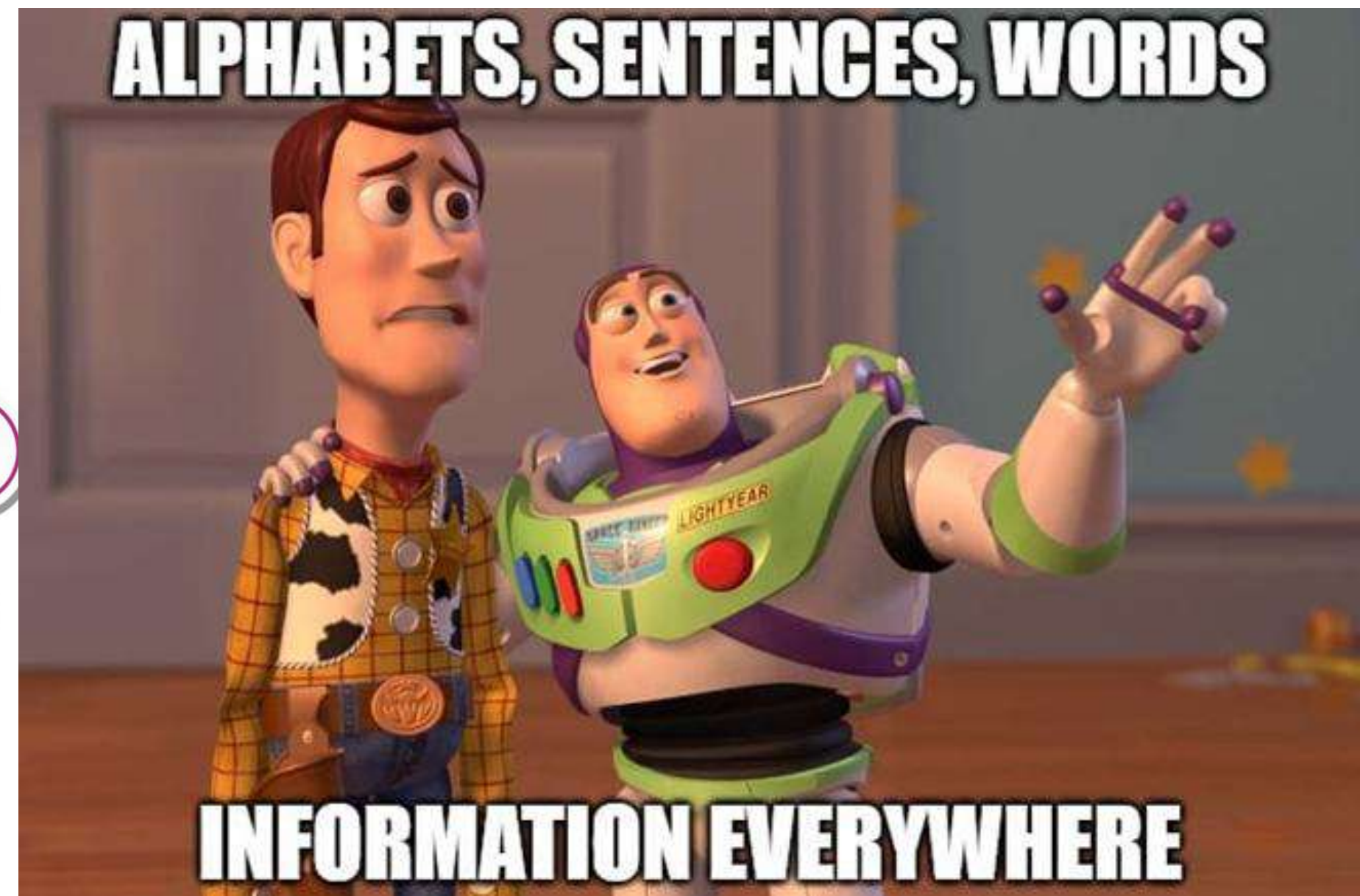
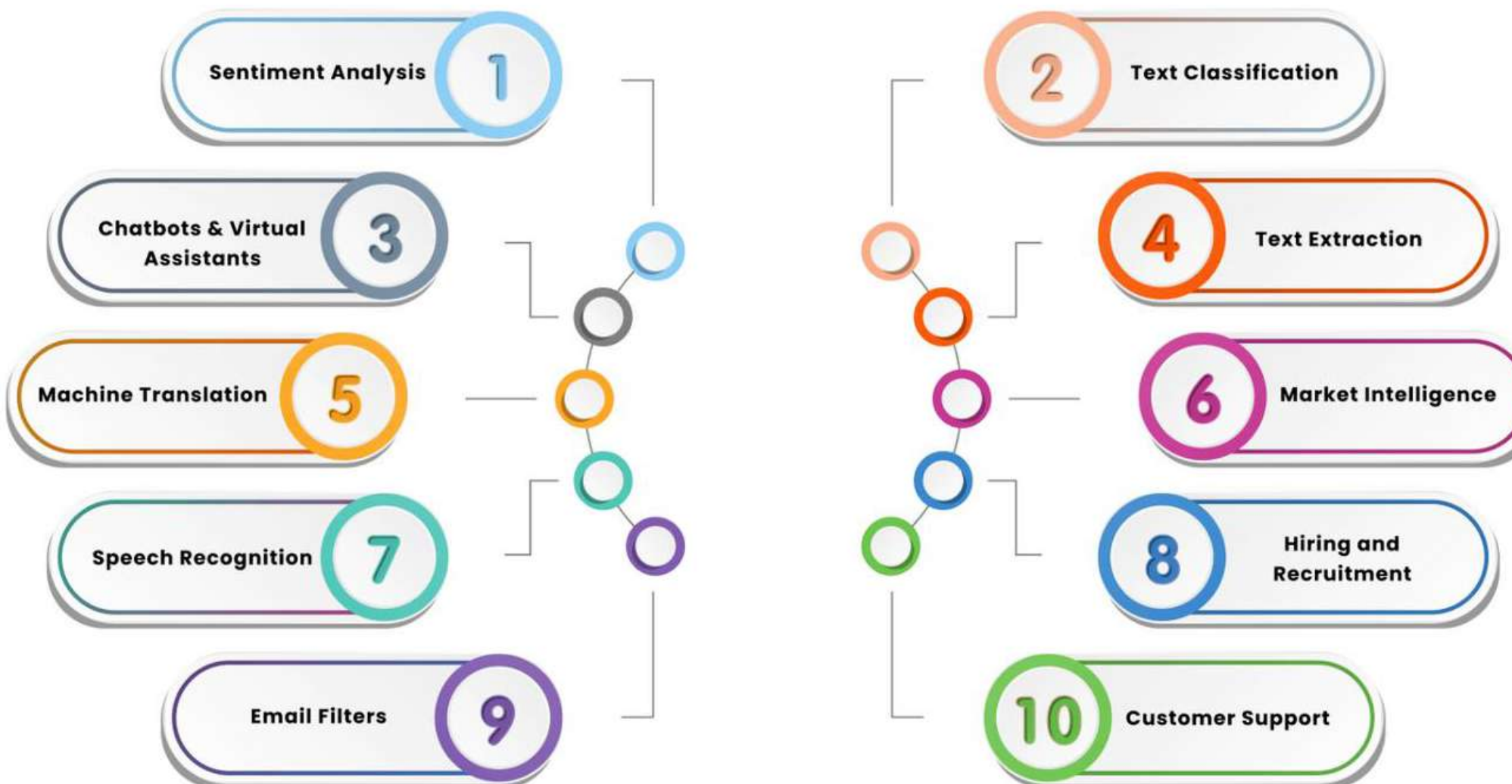


WHAT IS NLP ?

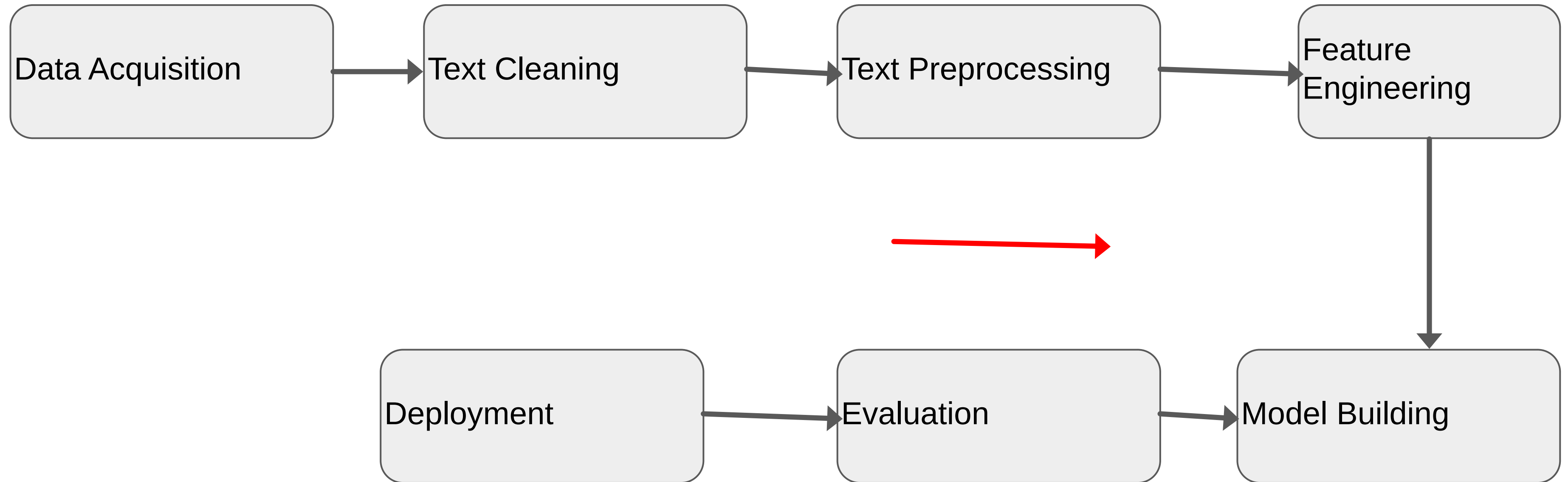
WHAT IS NLP ?

is referred to as NLP. It is a **subset** of AI that enables machines to comprehend and analyze **human languages**. Text or audio can be used to represent human languages.

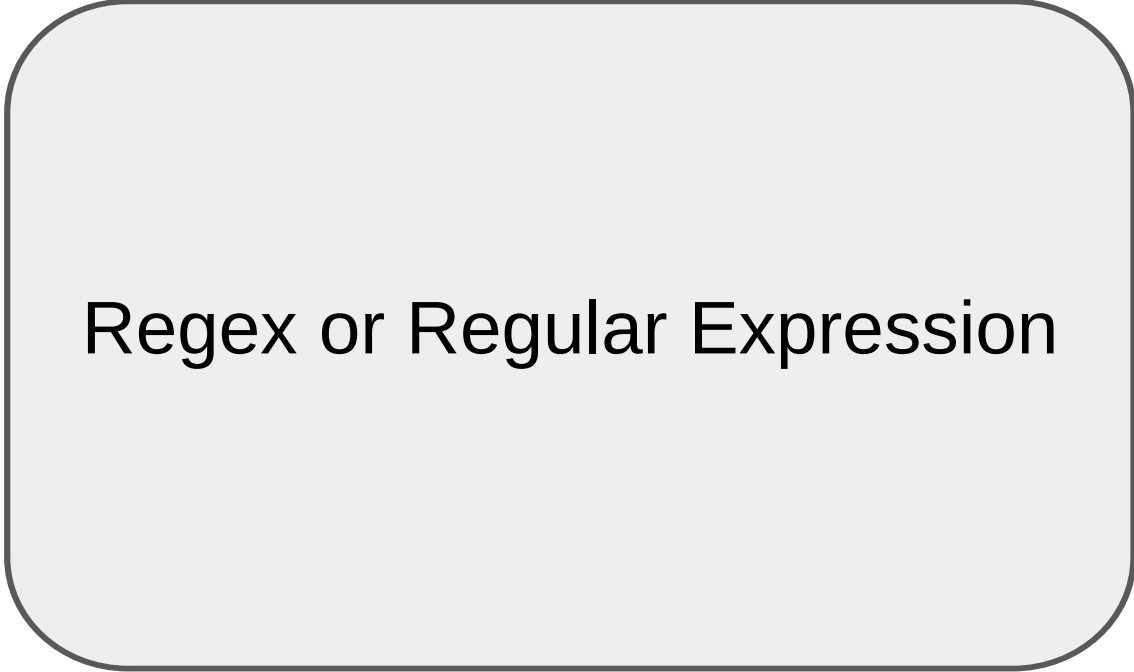
NLP applications



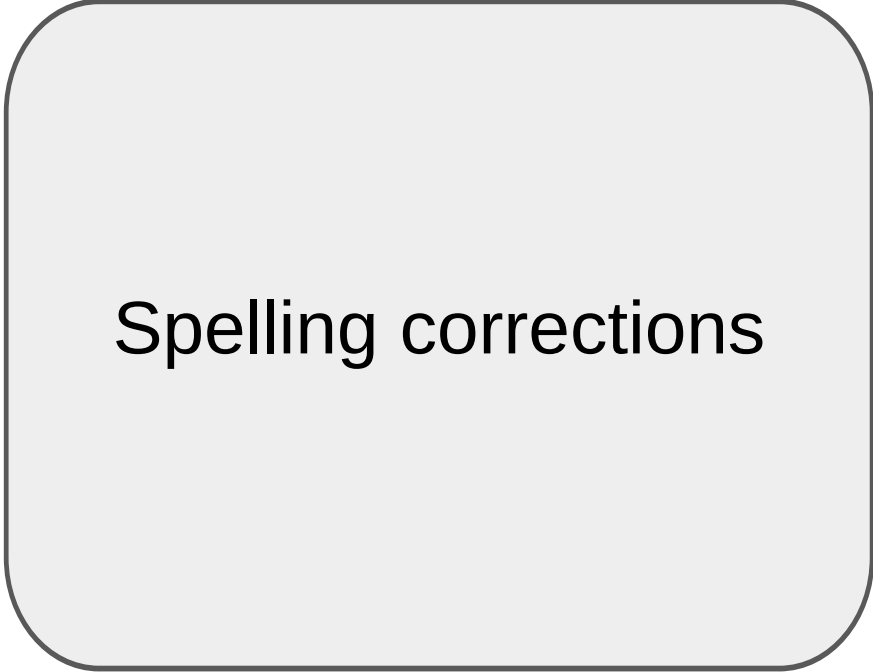
NLP Pipeline



Text Cleaning



Regex or Regular Expression

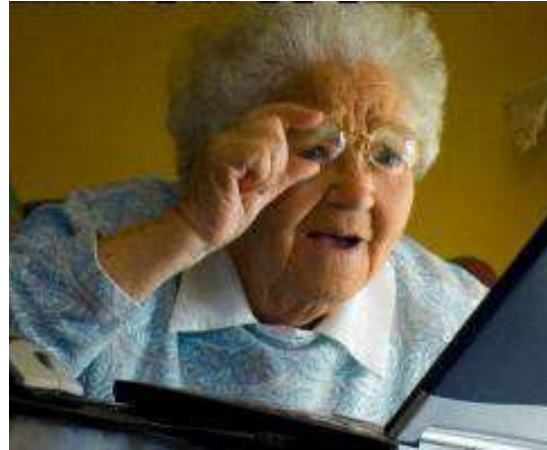


Spelling corrections

used for searching the string of specific patterns. Suppose our data contain phone number, email-id, and URL. we can find such text using the regular expression. After that either we can keep or remove such text patterns as per requirements.

Text Preprocessing

- Lowercasing
- Stop word removal
- Stemming or lemmatization
- Removing digit/punctuation
- POS tagging (adj, noun, adv)
- Named Entity Recognition (NER) (name, location,..)



**"Elon Musk founded SpaceX in
2002."**

**Named Entity Recognition
(NER)**

[Elon Musk: PERSON],
[SpaceX: ORG],
[2002: DATE].

**POS
tagging**

Elon/NNP (Proper noun)
Musk/NNP
founded/VBD
SpaceX/NNP
in/IN
2002/CD (Cardinal number)

Feature Engineering = Text Representation = Text Vectorization.

Our main agenda is to represent the text in the numeric vector in such a way that the ML algorithm can understand the text attribute.

Traditional Approach

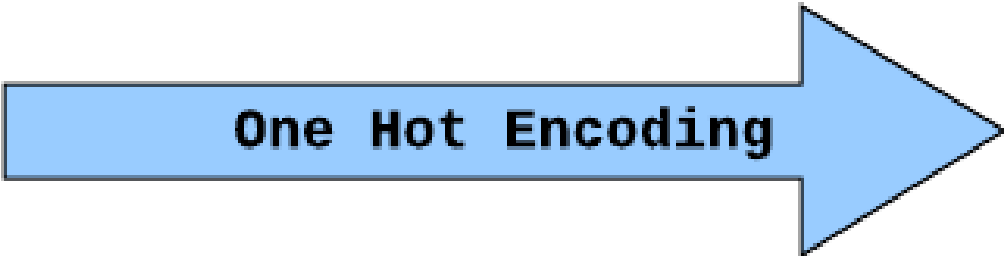
- One Hot Encoding



Traditional Approach

- One Hot Encoding

id	color
1	red
2	blue
3	green
4	blue



id	color_red	color_blue	color_green
1	1	0	0
2	0	1	0
3	0	0	1
4	0	1	0

Traditional Approach

- TF-IDF (Term Frequency – Inverse Document Frequency) 1972

Give more weight to rare words and less to common terms

$$TF(t, d) = \frac{(\text{Number of occurrences of term } t \text{ in document } d)}{(\text{Total number of terms in the document } d)}$$

$$IDF(t, D) = \log_e \frac{(\text{Total number of documents in the corpus})}{(\text{Number of documents with term } t \text{ in them})}$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$



GDG Carthage

Traditional Approach

- Neural Approach (Word embedding)

car =[0.8, 0.9, 0.9, 0.01, 0.75]

bike =[0.8, 0.7, 0.2, 0.01, 0.5]

not interpretable for humans



try to incorporate the contextual meaning of the words.

Word	car	bike
Road	0.8	0.8
Speed	0.9	0.7
Fuel	0.9	0.2
Animal	0.01	0.01
Price	0.75	0.5

How can we get these word embedding vectors?

How can we get these word embedding vectors?



Train our own embedding layer:

- **CBOW (Continuous Bag of Words)**
- **SkipGram**



Pre-Trained Word Embeddings

- **Word2vec by Google 2013**
- **GloVe by Stanford**
- **fasttext by Facebook**

Visualizing High-Dimensional Space:

https://www.youtube.com/watch?v=wvsE8jm1GzE&ab_chanel=GoogleforDevelopers

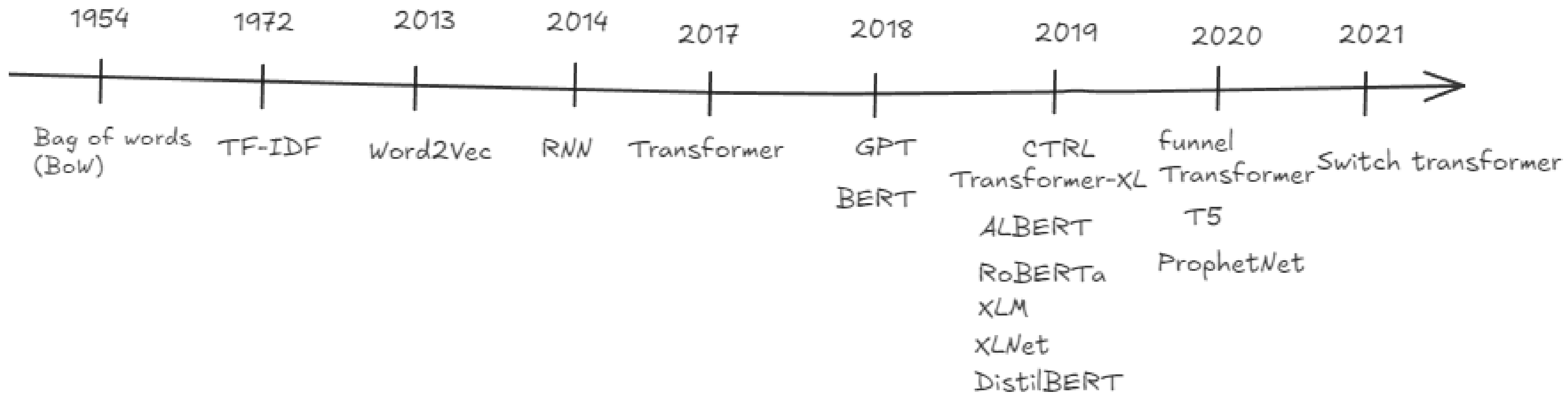
By Mohammed Arbi Nsibi

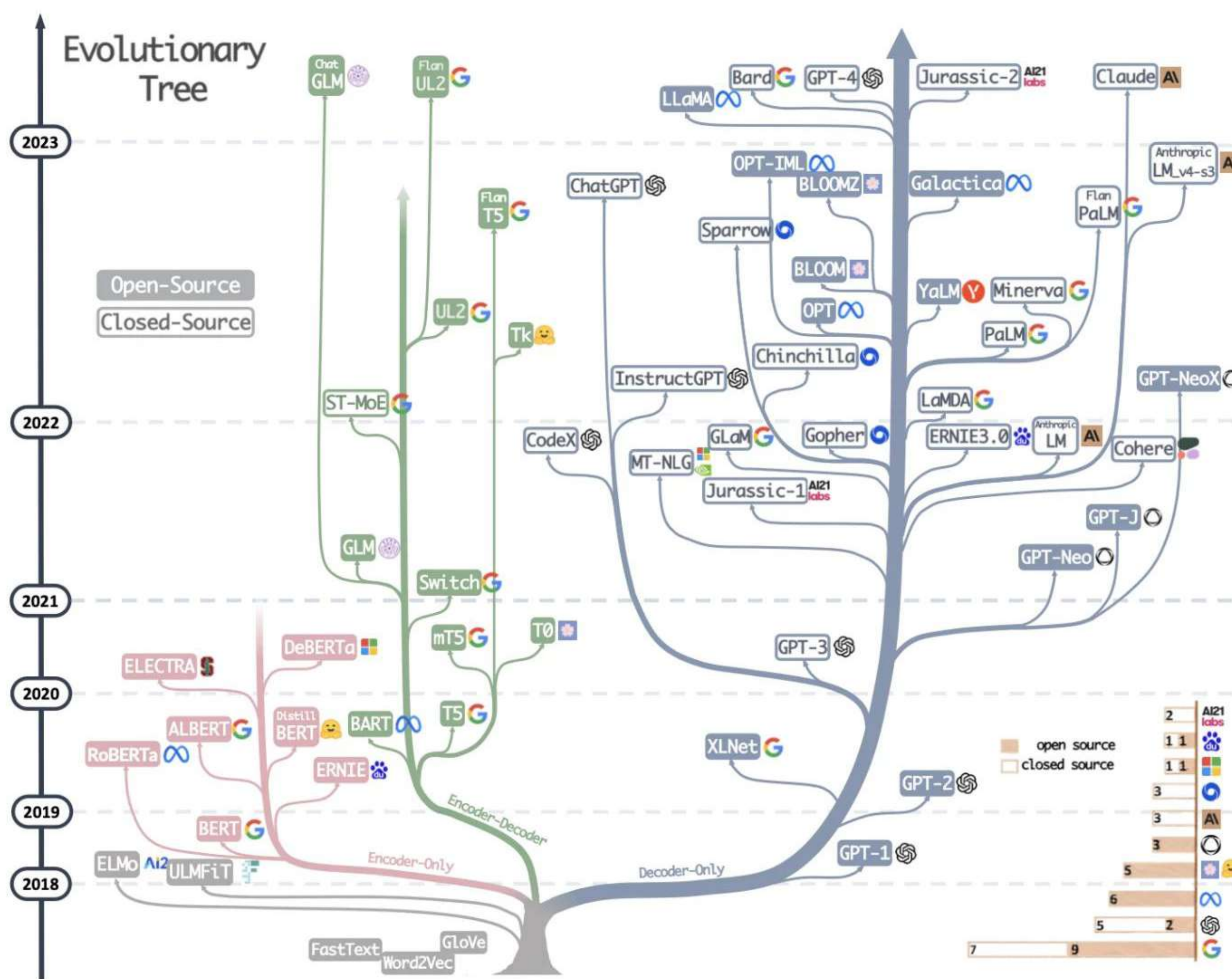
LET'S CODE 



NLP vs LLM







2	AI21 labs
1 1	du
1 1	
3	
3	AI
3	
5	
6	
5	2
7	9

The evolutionary tree of modern LLMs via <https://arxiv.org/abs/2304.13712>.

GPT-1

(June 2018)

6 years: What has changed?



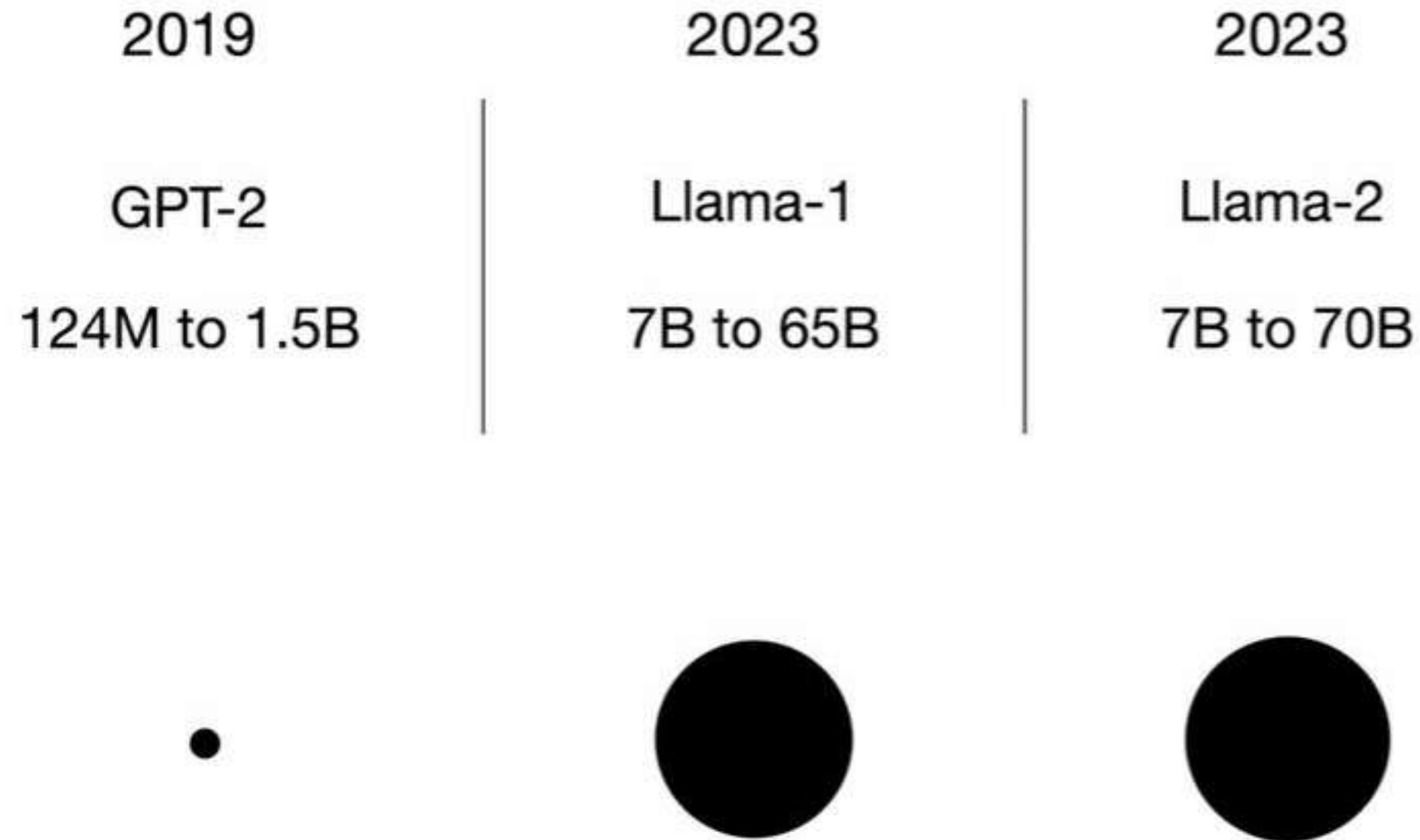
Llama 3.2

(September 2024)



GDG Carthage

Model size



GDG Carthage

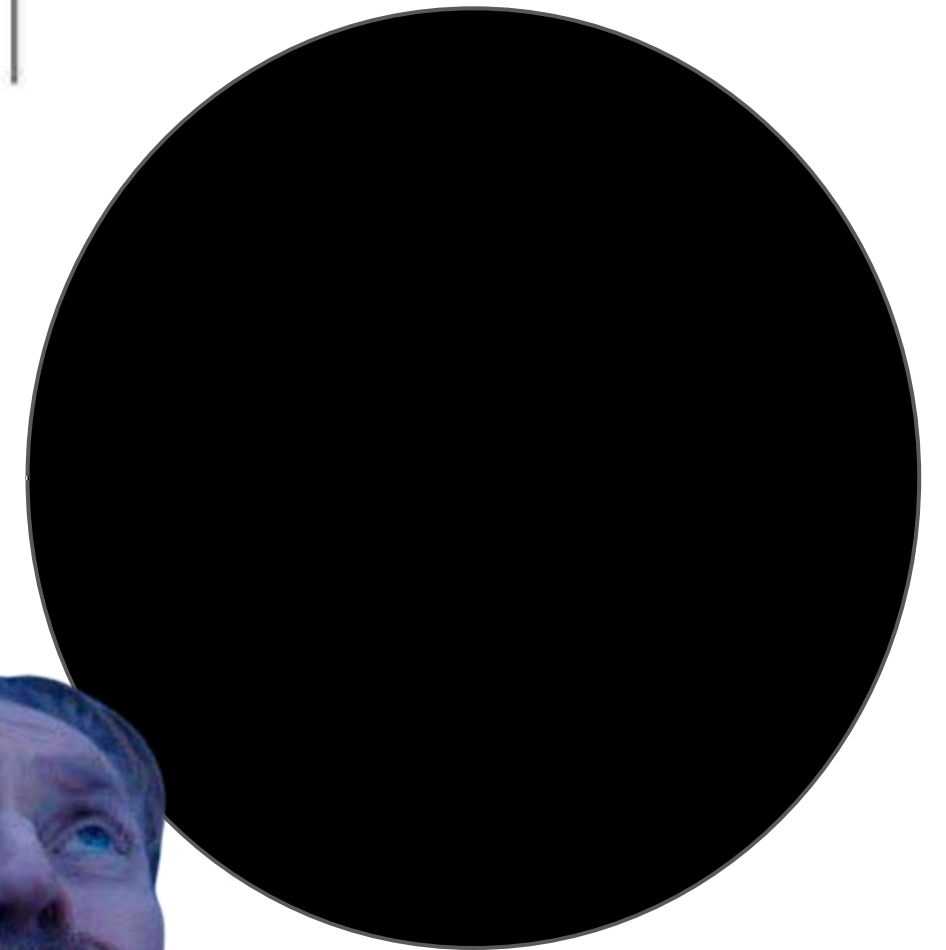
Model size

2019
GPT-2
124M to 1.5B

2023
Llama-1
7B to 65B

2023
Llama-2
7B to 70B

2024
Llama-3
8B to 405B



Dataset

2019

GPT-2

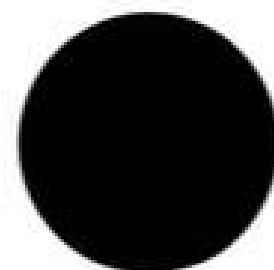
40B tokens



2023

Llama-1

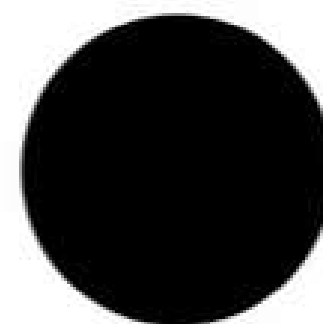
1.4T tokens



2023

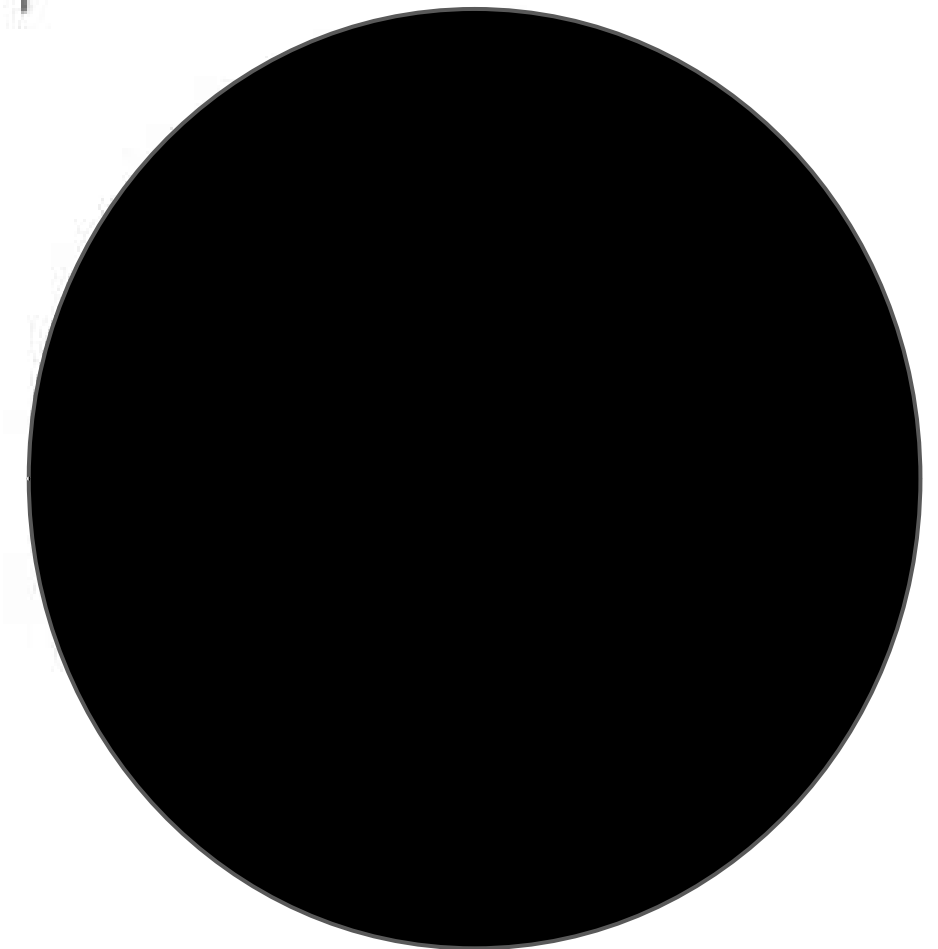
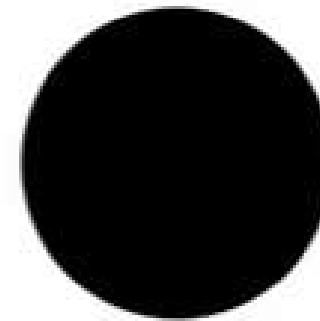
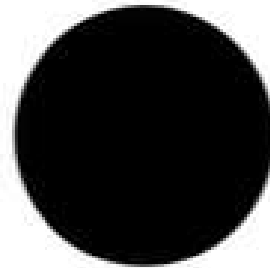
Llama-2

2T tokens



Dataset

2019	2023	2023	2024
GPT-2	Llama-1	Llama-2	Llama-3
40B tokens	1.4T tokens	2T tokens	15T tokens



GDG Carthage

"Attention is All You Need"



Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

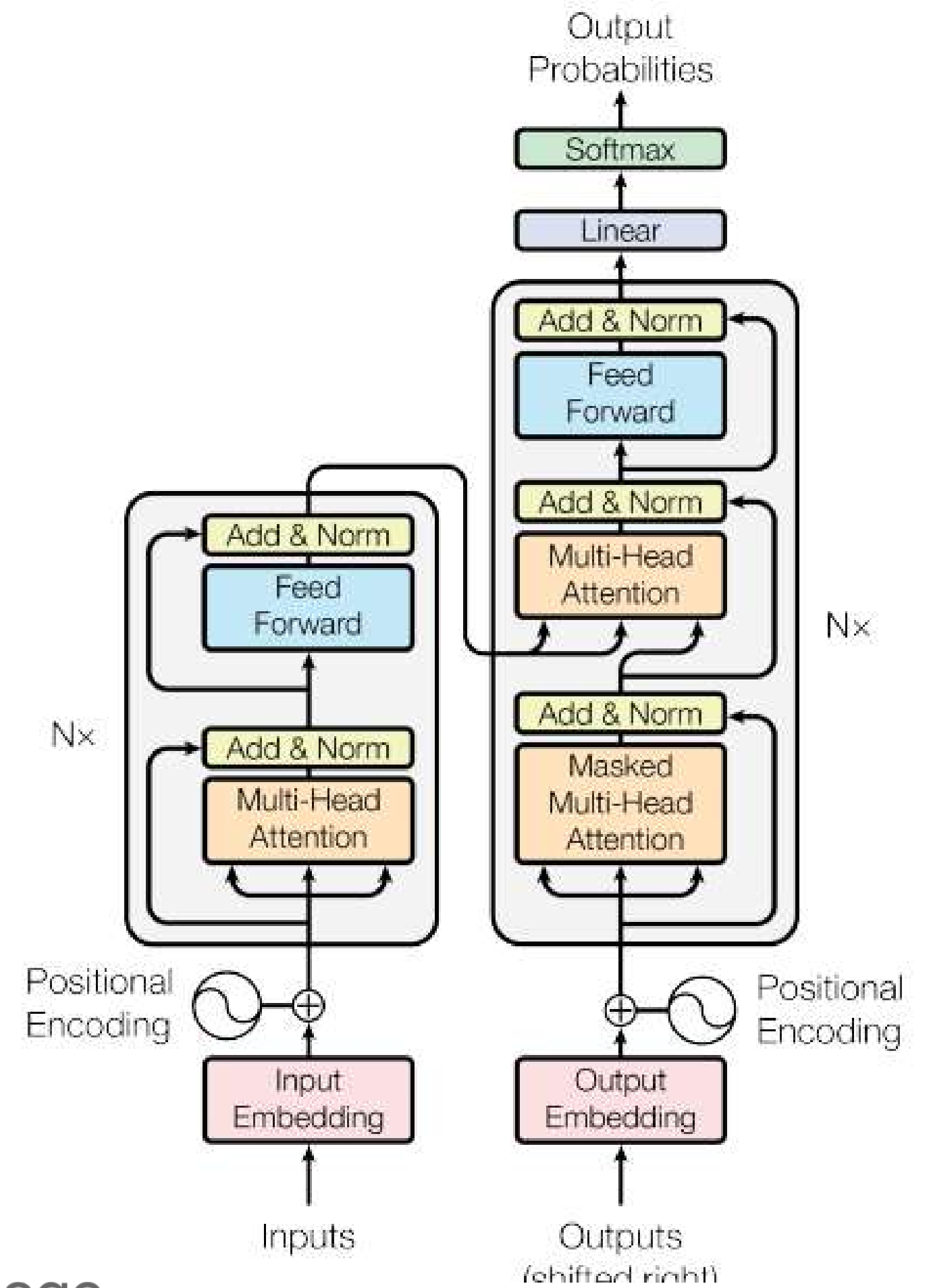
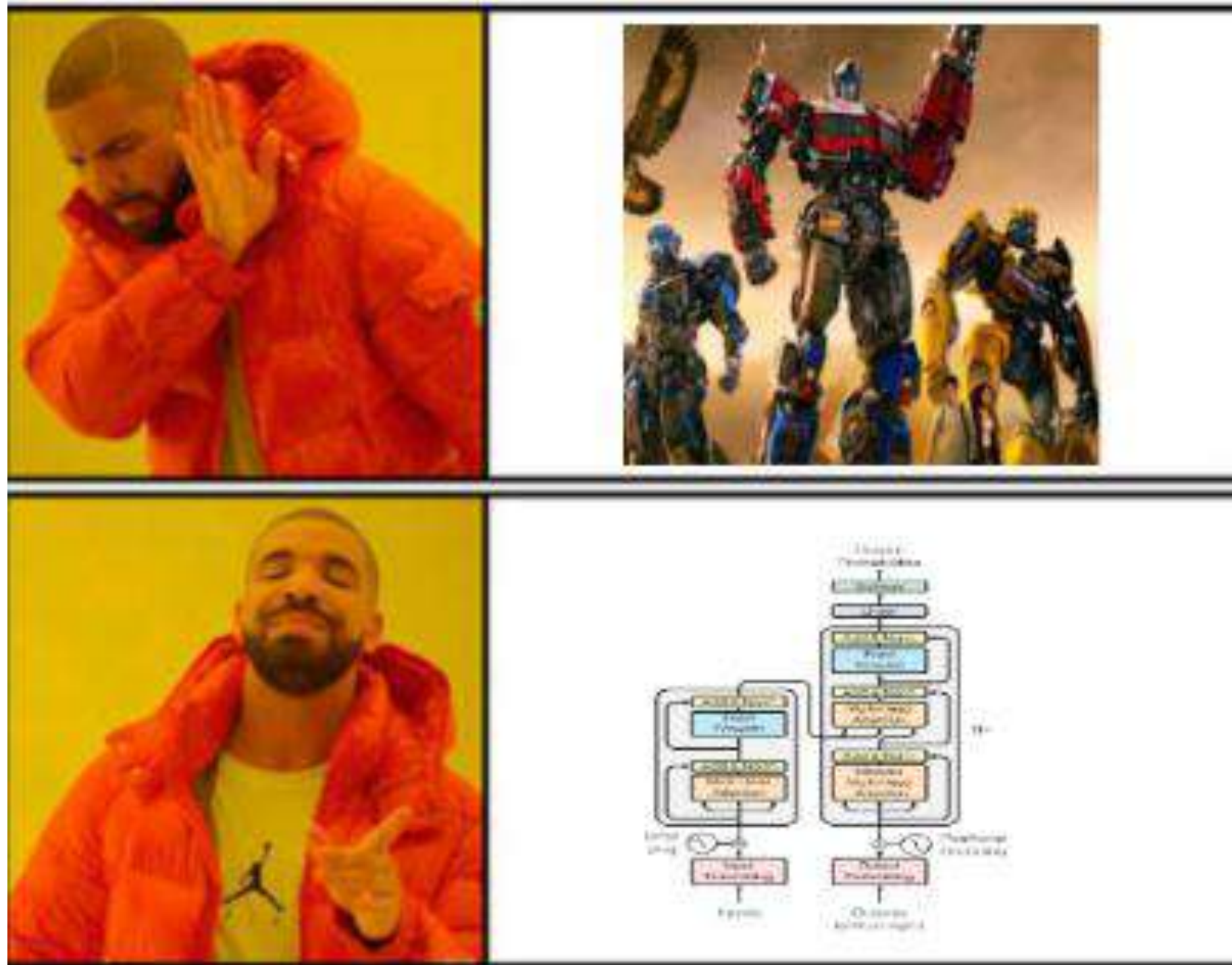
Lukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Transformers



X_1 for "I"

X_2 for "love"

X_3 for "NLP"

$$PE_{(pos\ 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE_{(pos\ 2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \end{bmatrix}$$

Position	PE (dim 0)	PE (dim 1)	PE (dim 2)	PE (dim 3)
0	$\sin(0) = 0$	$\cos(0) = 1$	$\sin(0) = 0$	$\cos(0) = 1$
1	$\sin(1/10000^0) \approx 0.8415$	$\cos(1/10000^0) \approx 0.5403$	$\sin(1/10000^0) \approx 0.8415$	$\cos(1/10000^0) \approx 0.5403$
2	$\sin(2/10000^0) \approx 0.9093$	$\cos(2/10000^0) \approx -0.4161$	$\sin(2/10000^0) \approx 0.9093$	$\cos(2/10000^0) \approx -0.4161$

$$E = X + PE$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

الوضع ريلاكس بلا صياح



Where to find pretrained LLMs ?



Hugging Face

Where to find pretrained LLMs ?

Models1,028,261

Filter by name

Full-text search

Sort: Trending

 openai/whisper-large-v3-turbo

Automatic Speech Recognition • Updated 1 day ago • 10k • 324

 black-forest-labs/FLUX.1-dev

Text-to-Image • Updated Aug 16 • 1.14M • 5.03k

 jasperai/Flux.1-dev-Controlnet-Upscaler

Image-to-Image • Updated 3 days ago • 9.86k • 244

 allenai/Molmo-7B-D-0924

Image-Text-to-Text • Updated 1 day ago • 14.5k • 273

 meta-llama/Llama-3.2-11B-Vision-Instruct

Image-Text-to-Text • Updated 4 days ago • 139k • 479

 nvidia/NVLM-D-72B

Image-Text-to-Text • Updated about 18 hours ago • 860 • 242

 meta-llama/Llama-3.2-1B

Text Generation • Updated 3 days ago • 61.2k • 299

 openbmb/MiniCPM-Embedding

Feature Extraction • Updated 2 days ago • 130k • 204

Datasets222,500

Filter by name

Full-text search

Sort: Trending

 google/frames-benchmark

Viewer • Updated about 17 hours ago • 824 • 562 • 122

 fka/awesome-chatgpt-prompts

Viewer • Updated Sep 3 • 170 • 8.36k • 5.82k

 FBK-MT/mosel

Viewer • Updated 5 days ago • 51.1M • 21 • 42

 migtissera/Synthia-v1.5-I

Viewer • Updated 8 days ago • 20.7k • 99 • 39

 openai/MMMLU

Viewer • Updated 4 days ago • 393k • 5.33k • 374

 HackerNoon/where-startups-trend

Preview • Updated 7 days ago • 19 • 36

 argilla/FinePersonas-v0.1

Viewer • Updated 19 days ago • 21.1M • 371 • 304

 k-mktr/improved-flux-prompts-photoreal-portrait

Viewer • Updated 4 days ago • 20k • 54 • 62

Model types

Text-to-text

Text-to-image

Text-to-video

Sentence-similarity

 Image-Text-to-Text

 Visual Question Answering

 Document Question Answering

 Video-Text-to-Text

 Any-to-Any

Computer Vision

 Depth Estimation

 Image Classification

 Object Detection

 Image Segmentation

 Text-to-Image

 Image-to-Text

 Image-to-Image

 Image-to-Video

 Unconditional Image Generation

 Video Classification

 Text-to-Video

 Zero-Shot Image Classification

 Mask Generation

 Zero-Shot Object Detection

 Text-to-3D

 Image-to-3D

 Image Feature Extraction

 Keypoint Detection

Natural Language Processing

 Text Classification

 Token Classification

 Table Question Answering

 Question Answering

 Zero-Shot Classification

 Translation

 Summarization

 Feature Extraction

 Text Generation

 Text2Text Generation

 Fill-Mask

 Sentence Similarity

Audio

 Text-to-Speech

 Text-to-Audio

 Automatic Speech Recognition

 Audio-to-Audio

 Audio Classification

 Voice Activity Detection

How to create my own application

```
import gradio as gr
from transformers import pipeline

pipe = pipeline("sentiment-analysis")

def predict(new_input):
    out = pipe(new_input)
    out = out[0]["label"]
    return out

gr.Interface(predict, inputs=["text"], outputs=["text"]).launch()
```

<https://huggingface.co/learn/nlp-course/chapter1/3#working-with-pipelines>

LET'S CODE



By Mohammed Arbi Nsibi

Hallucinations

- The model is not trained on enough data.
- The model is trained on noisy or dirty data.
- The model is not given enough context .
- The model is not given enough constraints.

LLM AFTER TRAINING
ON 90% OF THE INTERNET...



Hallucinations



Is 9677 a prime number?



No, 9677 is not a prime number.
It can be factored into 13 and 745, as $9677 = 13 \times 745$.

} incorrect assertion
} snowballed hallucination

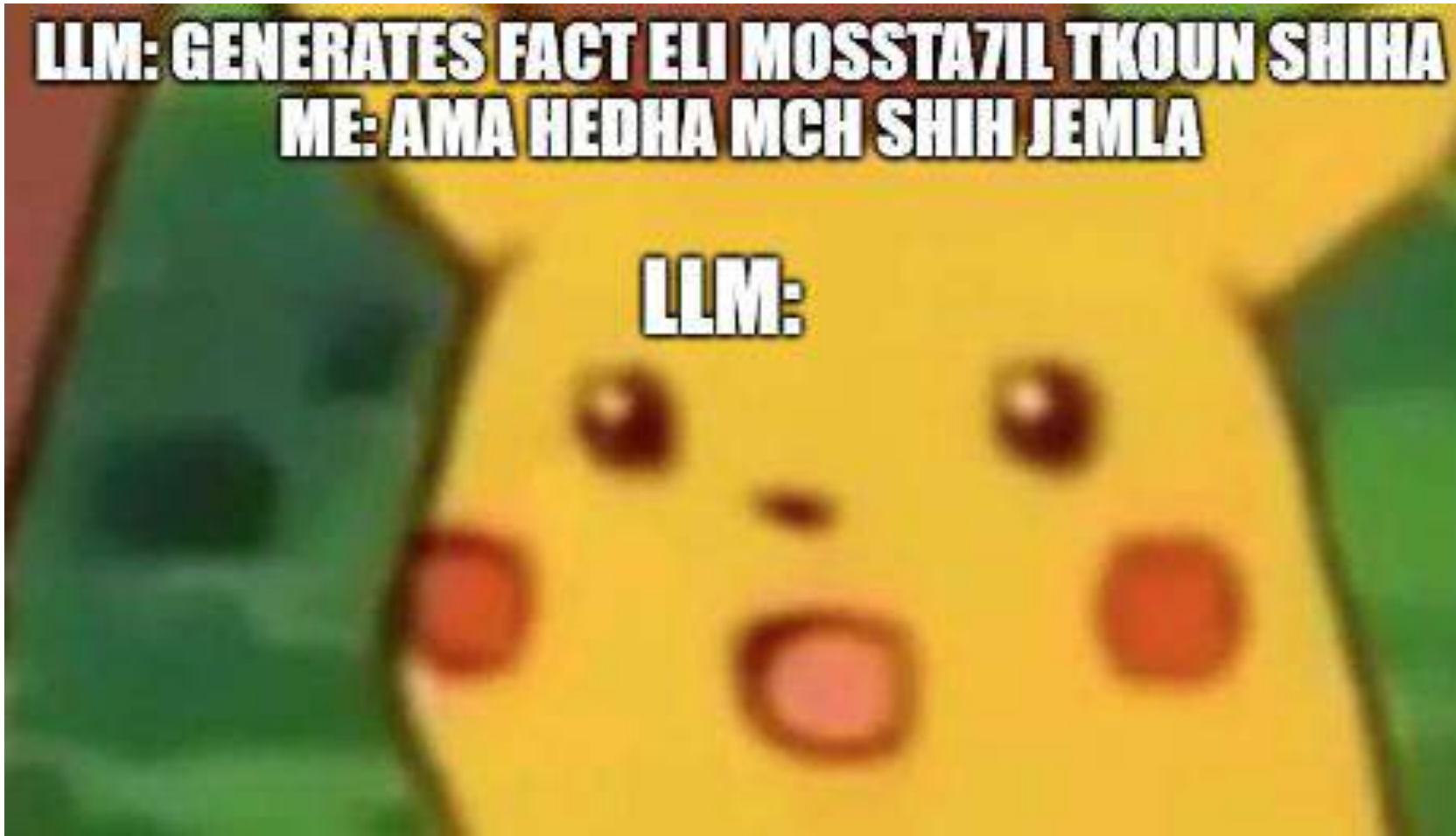
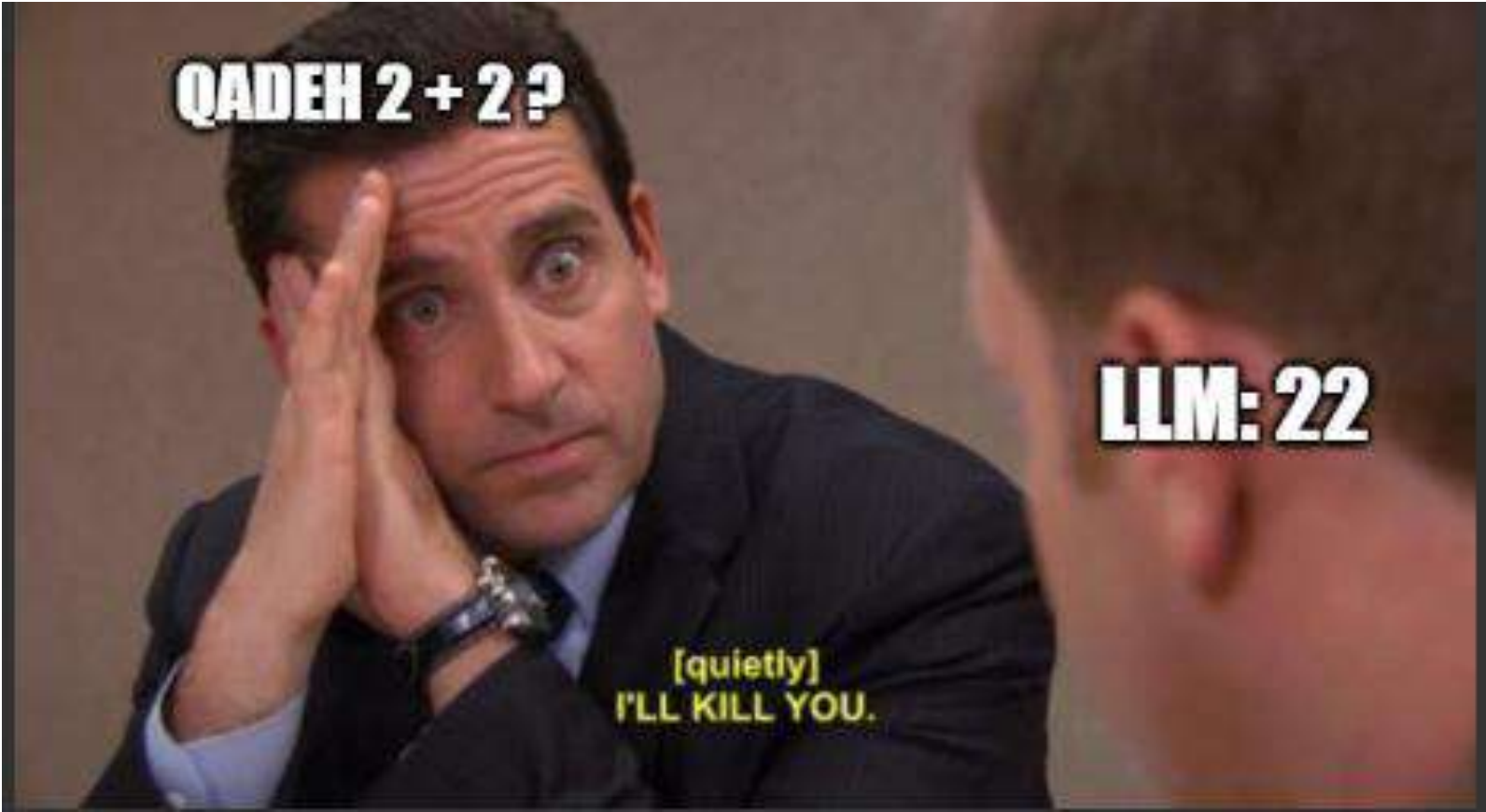


Is 9677 divisible by 13?

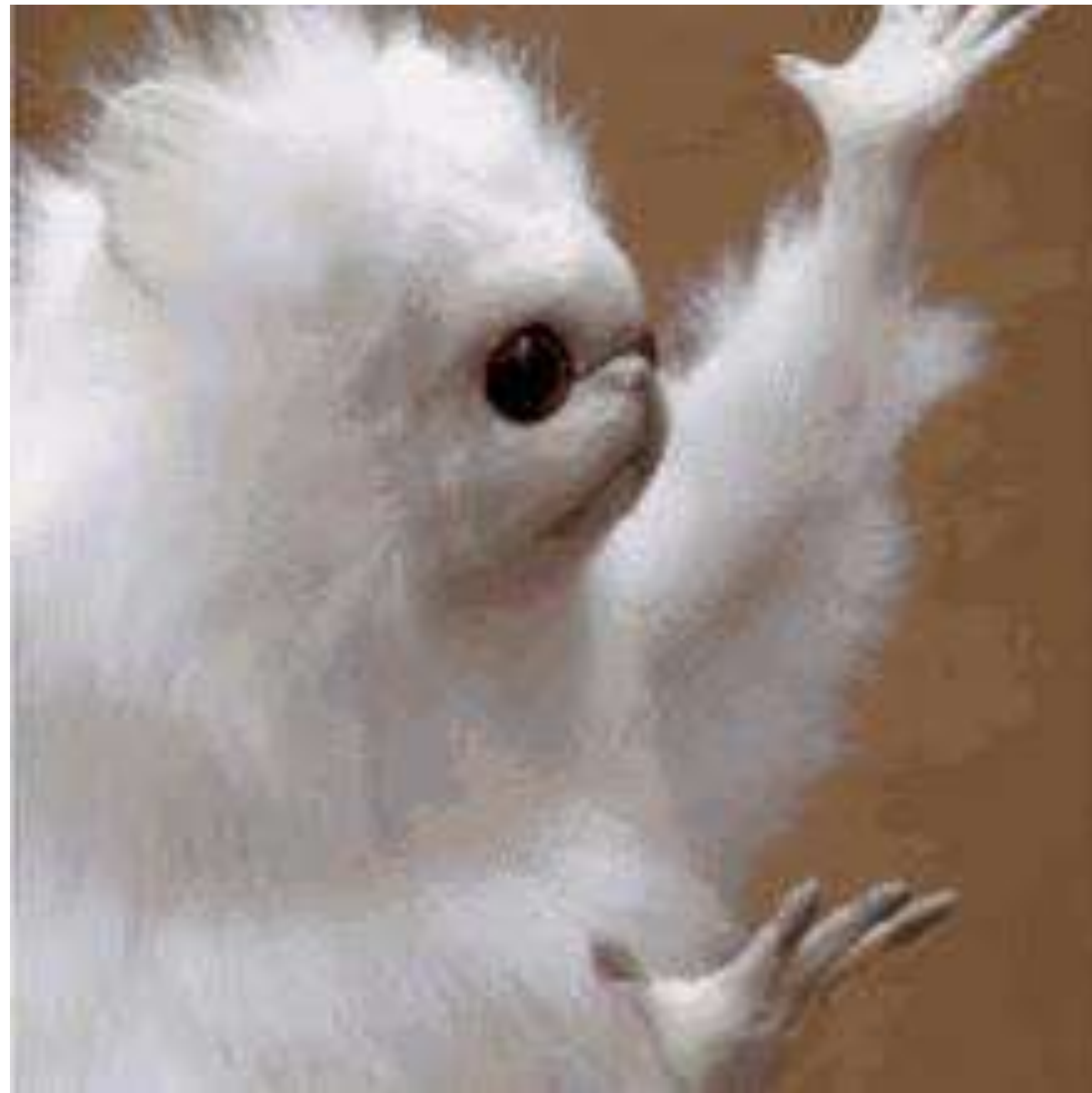


No

in a separate session, GPT-4 recognizes its claim as incorrect!

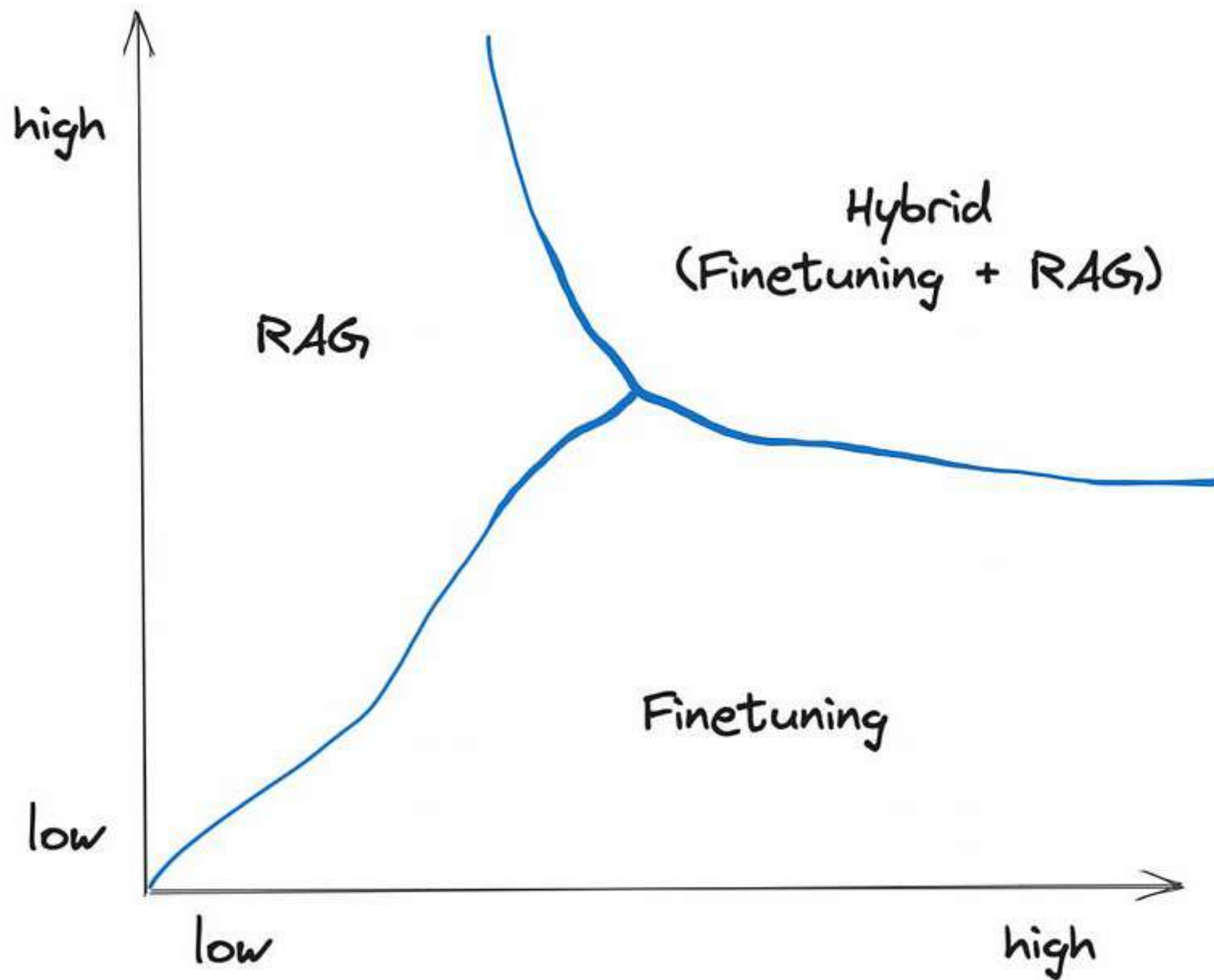


Solutions ?



RAG / Fine-tuning

external knowledge
required

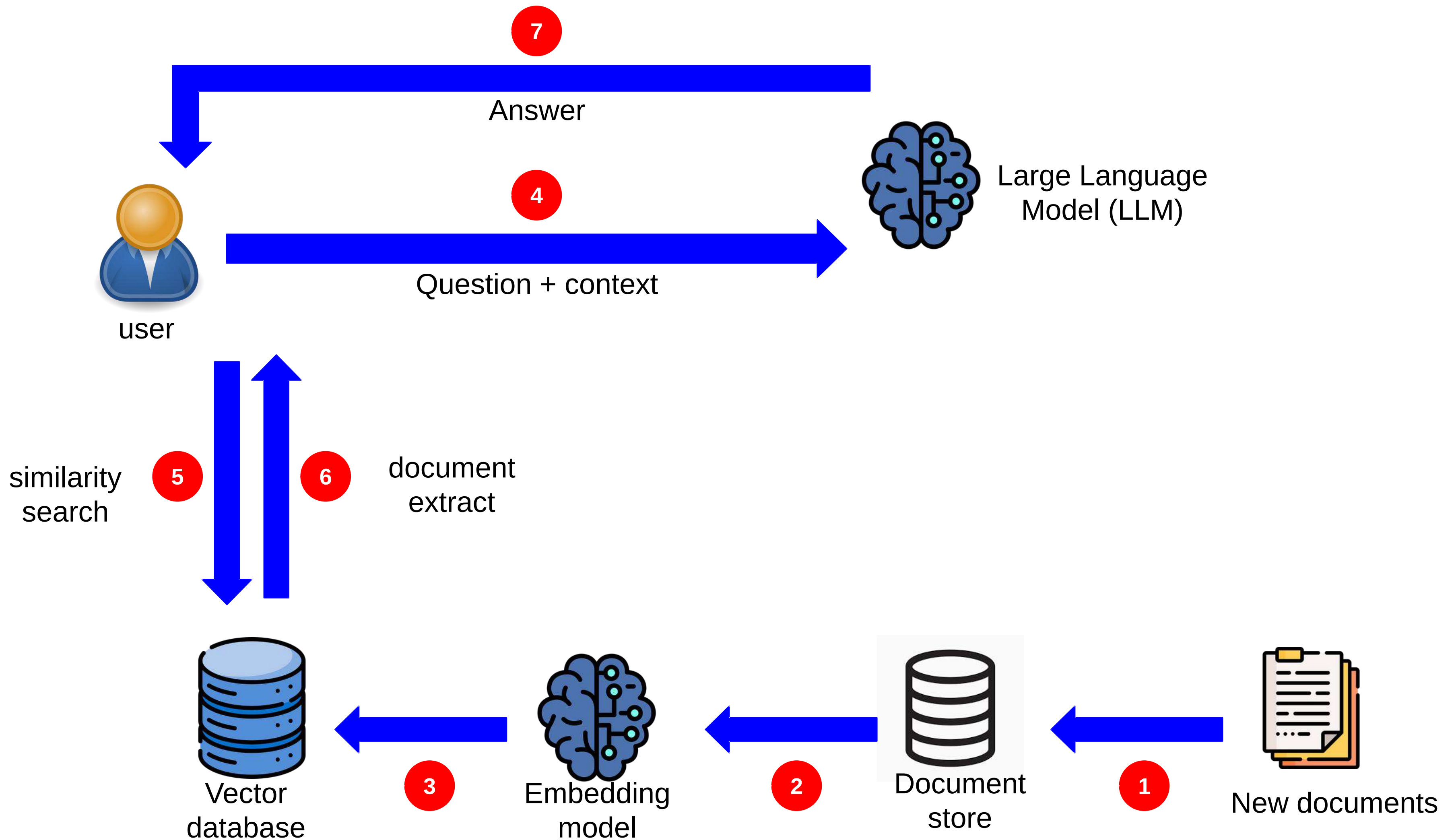


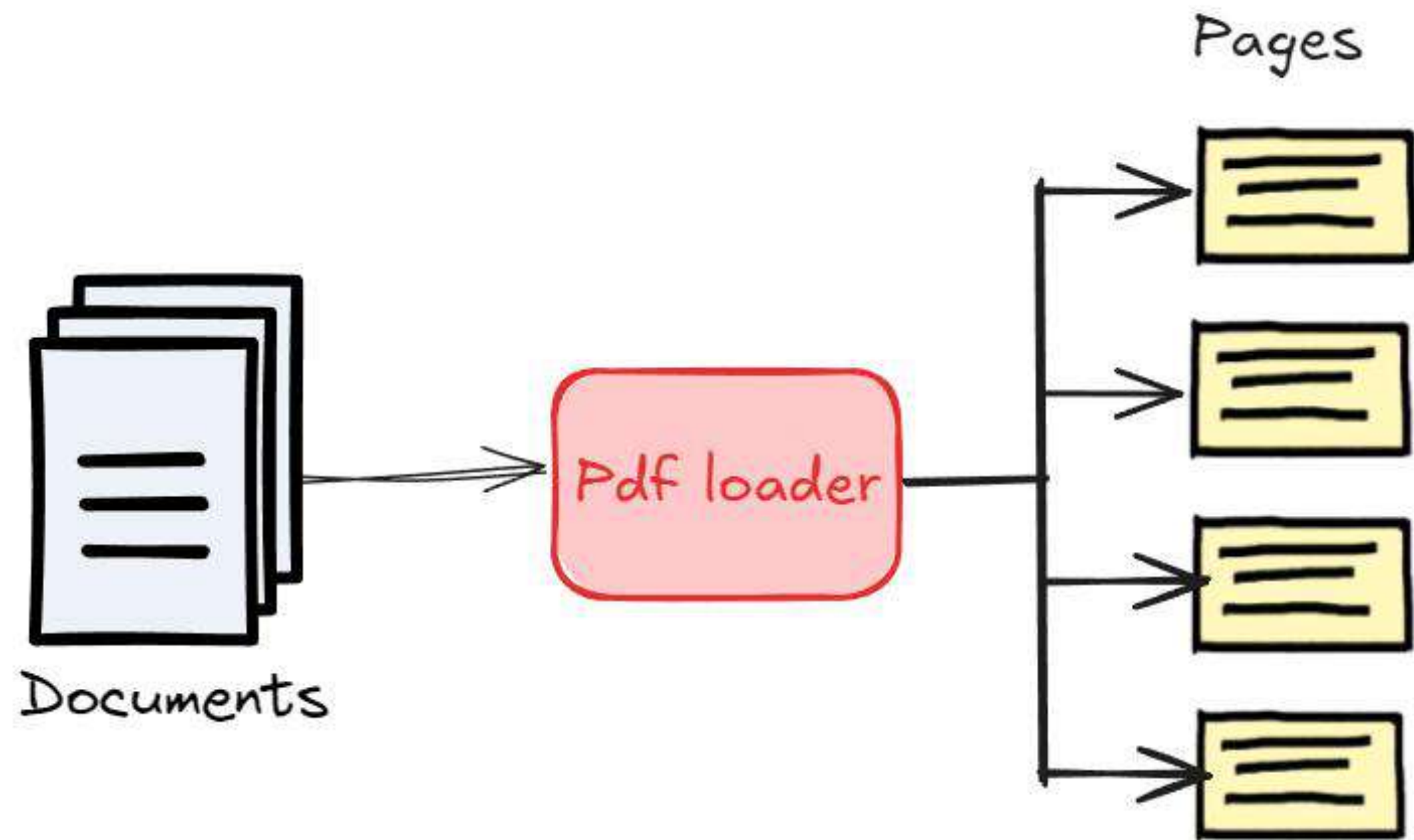
model adaptation required
(e.g. behaviour/
writing style/
vocabulary)

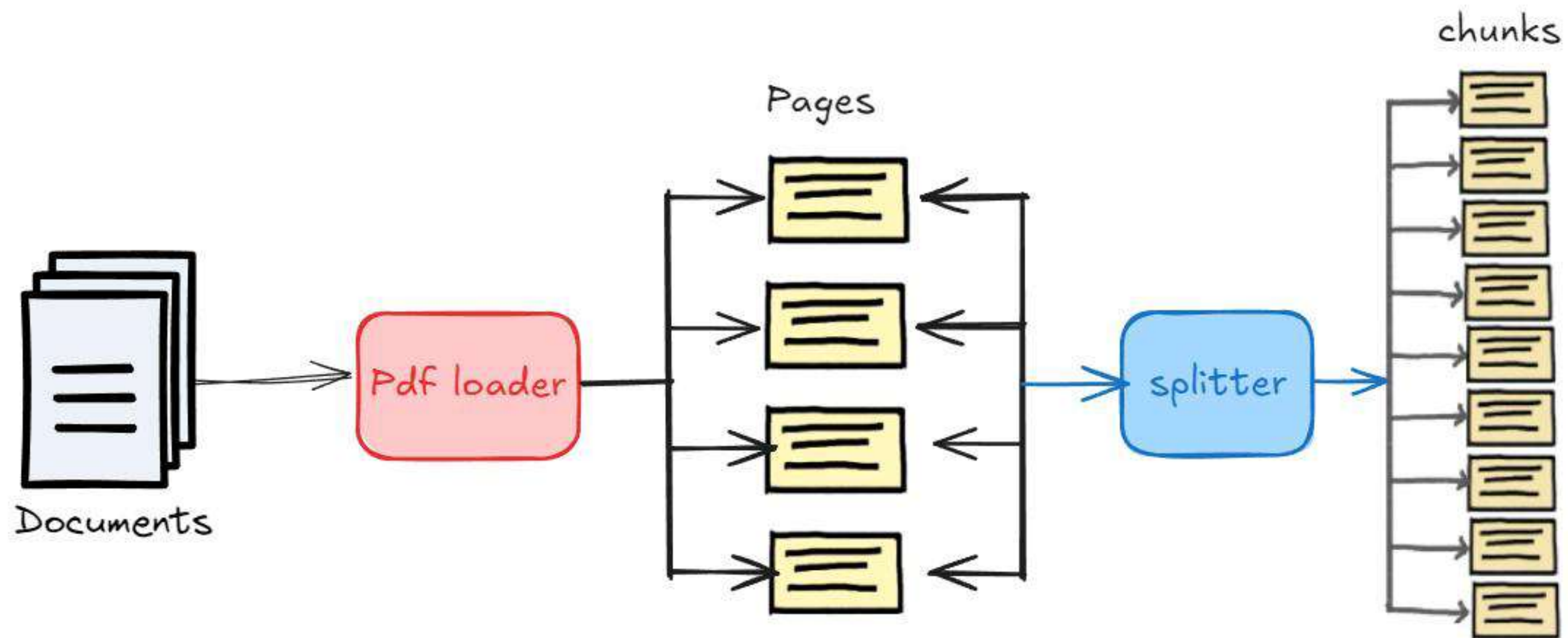
**But wait..
WTF is RAG ?**



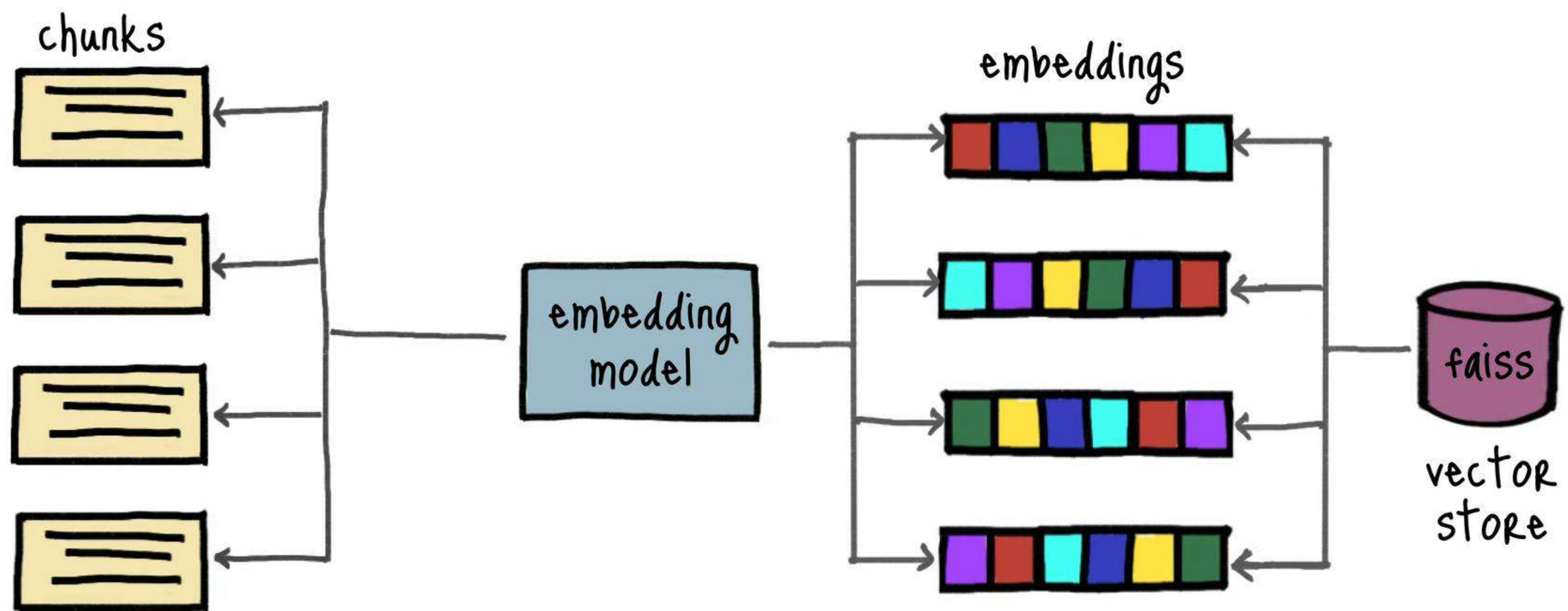
Wait a minute, who are you?

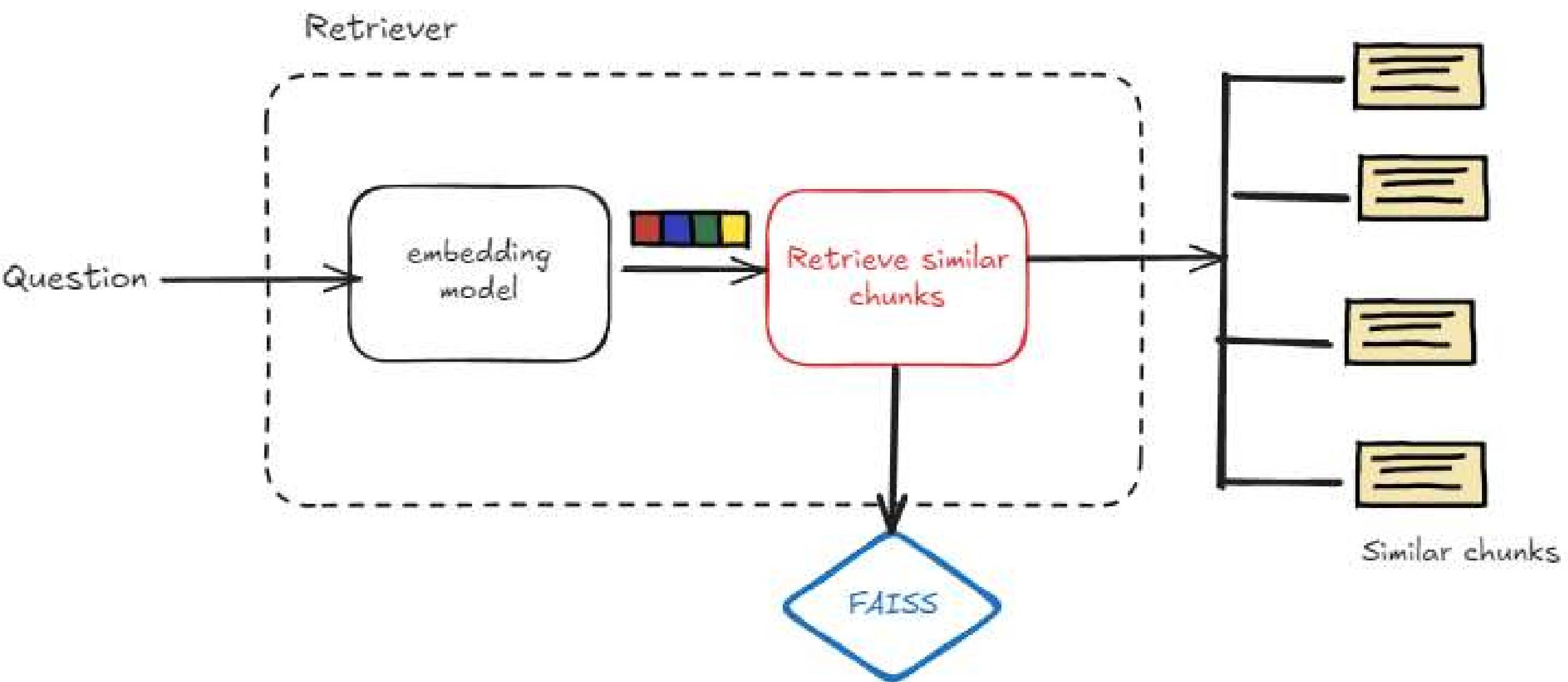




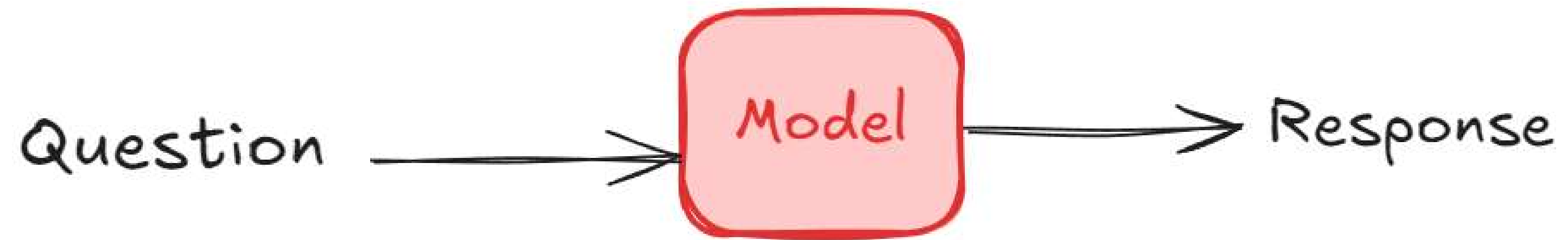


By Mohammed Arbi Nsibi

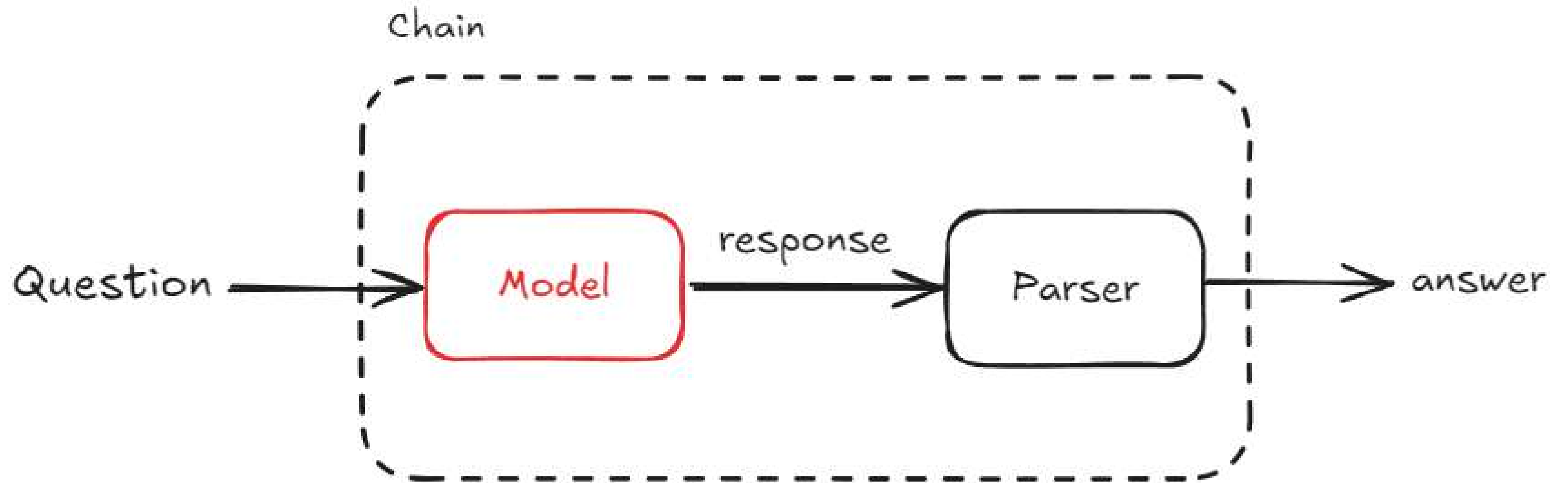




By Mohammed Arbi Nsibi

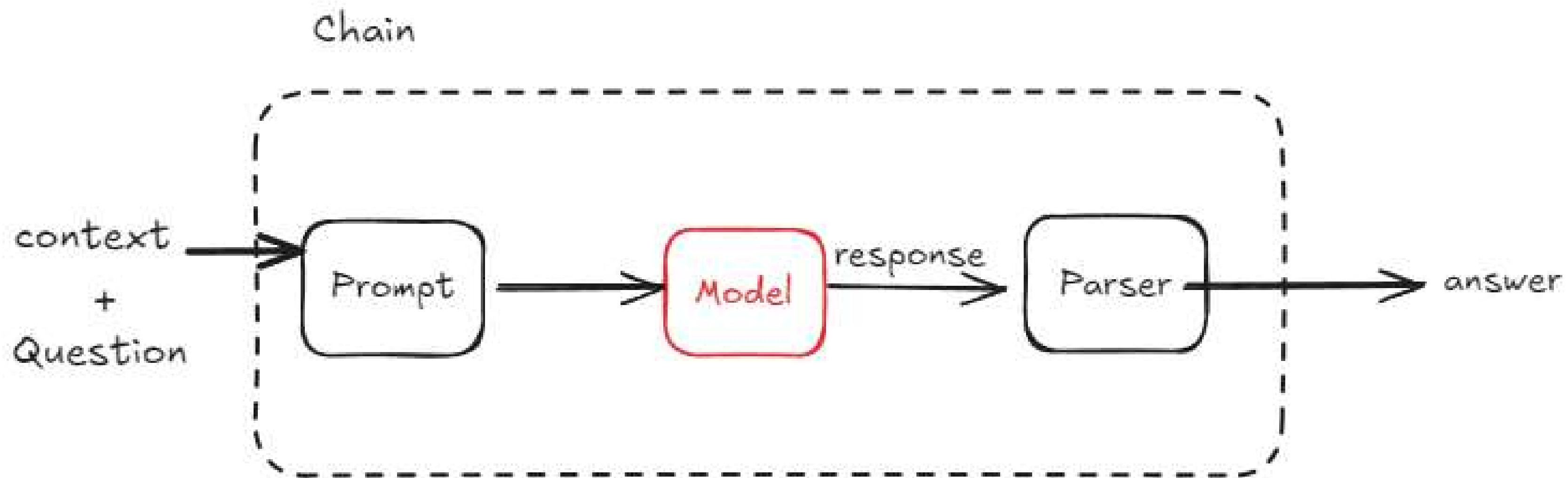



```
AIMessage(content='As of my last update in April 2023, Joe Biden is the President of the United States. He took office on January 20, 2021,
```

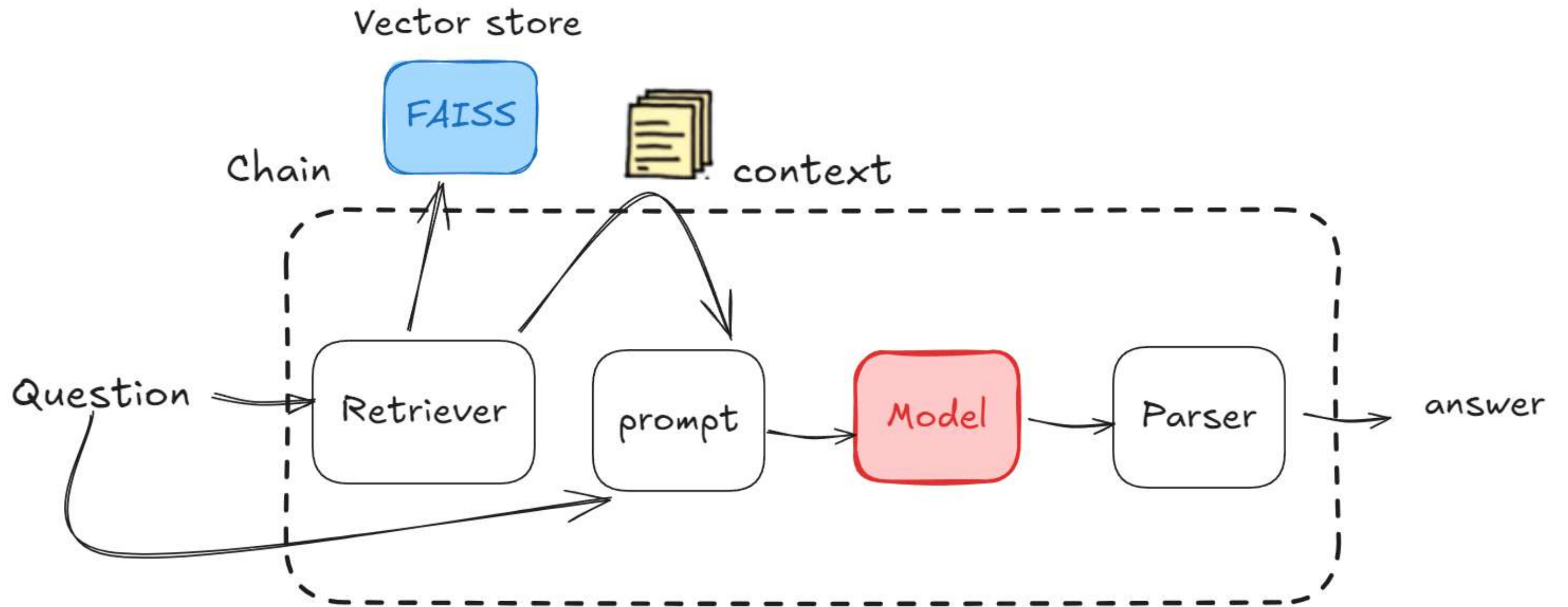


```
'As of my last update in April 2023, Joe Biden is the President of the United States. He took office on January 20, 2021,
```





By Mohammed Arbi Nsibi



RAG (Retrieval-augmented generation)



By Mohammed Arbi Nsibi

always has been

wait so creating chatbots is that easy ?





Q&A

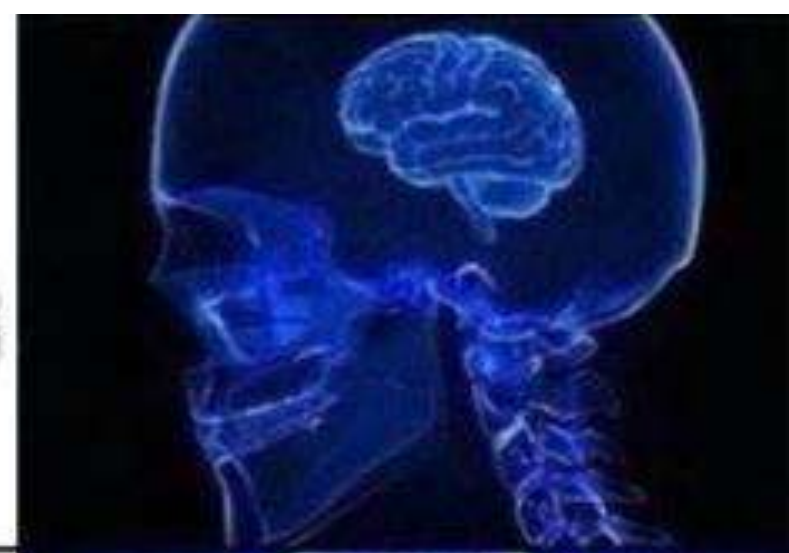
THANK YOU for you attention!!

**USING AI
FOR CALCULATIONS**

**USING AI TO
WRITE ESSAYS**

**USING AI TO
GENERATE CODE**

**USING AI
TO GENERATE
MEMES ABOUT AI**



QUIZ TIME 😁



Cost to pretrain a 685B parameter LLM
(not including failed runs, hyperparameter tuning, or personnel costs)

Training Costs	Pre-Training	Context Extension	Post-Training	Total
in H800 GPU Hours	2664K	119K	5K	2788K
in USD	\$5.328M	\$0.238M	\$0.01M	\$5.576M

Table 1 | Training costs of DeepSeek-V3, assuming the rental price of H800 is \$2 per GPU hour.

\$5 million!

https://github.com/deepseek-ai/DeepSeek-V3/blob/main/DeepSeek_V3.pdf

It makes me appreciate all these openly available models!

Math:

- The total number of GPU hours needed is 184,320 hours.
- The cost of running one A100 instance per hour is approximately \$33.
- Each instance has 8 A100 GPUs.

That's $184320 / 8 * 33 = \$760,000$